



**IMPACTS OF DATA-DRIVEN DEMAND FORECASTING IN REDUCING FOOD
WASTE AND CO2 EMISSIONS IN CAMPUS RESTAURANTS:
A CASE STUDY**

Lappeenranta–Lahti University of Technology LUT

Master's thesis in Environmental Technology

Master's Programme in Circular Economy

2025

Sara Mattila

Examiners: Professor Ville Uusitalo (D.Sc. Tech)

Junior researcher Jubeen Sharbaf (M.Sc. Tech)

ABSTRACT

Lappeenranta–Lahti University of Technology LUT
LUT School of Energy Systems
Environmental Technology

Sara Mattila

Impacts of data-driven demand forecasting in reducing food waste and CO₂ emissions in campus restaurants: A case study

Master's thesis
2025

60 pages, 8 figures, 6 tables and 3 appendices

Examiners: Professor Ville Uusitalo (D.Sc. Tech) and Junior researcher Jubeen Sharbaf (M.Sc. Tech)

Keywords: food waste reduction, demand forecasting, predictive models, CO₂ emissions, campus restaurants, machine learning, institutional food services, institutional climate action

Food waste in public food services is a growing global concern, contributing to unnecessary greenhouse gas emissions and inefficient use of natural resources. In university campus restaurants, overproduction is a common issue due to fluctuating diner numbers and the lack of reliable forecasting software. Solving this problem is important for helping universities take responsibility for the environment and lead by example in institutional climate action.

This study investigates the design, implementation and use of machine learning-based demand forecasting software created to reduce overproduction as part of the Waste Watch project at LUT University. The case study focuses on Kampusravintolat Oy, a university-owned food service company. The aim was to assess the impact of a data-driven demand forecasting software on food waste levels, CO₂ emissions, and restaurant operations in a real-world setting.

The study applied an action research approach, combining waste and emissions data analysis, software functionality evaluation, and discussions with kitchen staff and supervisors. During the pilot, the forecasting tool reduced service waste by an average of 16%, reaching up to 46% in individual months. This corresponds to an estimated annual reduction of 30.5 tons of CO₂-equivalent emissions. The tool also supported production planning, cost-efficiency, and staff engagement in climate action.

The results indicate that demand forecasting can serve as a scalable sustainability strategy for food services. By preventing overproduction at the source, data-driven forecasting helps reduce emissions, prevent food waste, and advance institutional climate goals. The study offers practical insights for food service providers, sustainability managers, and researchers developing machine learning-based tools for emission reduction.

TIIVISTELMÄ

Lappeenrannan–Lahden teknillinen yliopisto LUT
LUTin energiajärjestelmien tiedekunta
Ympäristötekniikka

Sara Mattila

Datapohjaisen kysynnän ennakkoinnin vaikutukset ruokahävikin ja CO₂-päästöjen vähentämisessä kampusravintoloissa: Tapaustutkimus

Ympäristötekniikan diplomityö

2025

60 sivua, 8 kuvaa, 6 taulukkoa ja 3 liitettä

Tarkastajat: Professori Ville Uusitalo (tekniikan tohtori) and Nuorempi Tutkija Jubeen Sharbaf (diplomi-insinööri)

Avainsanat: ruokahävikin vähentäminen, kysynnän ennakkointi, ennustemallit, CO₂-päästöt, kampusravintolat, koneoppiminen, julkiset ruokapalvelut, ilmastotoimet organisaatioissa

Ruokahävikki julkisissa ruokapalveluissa on merkittävä globaali ongelma, joka aiheuttaa tarpeettomia kasvihuonekaasupäästöjä ja luonnonvarojen tehotonta käyttöä. Yliopistojen kampusravintoloissa ylituotantoa syntyy erityisesti ruokailijamäärien vaihtelun ja ennusteohjelmistojen puutteen vuoksi. Ongelman ratkaiseminen on tärkeää, jotta yliopistot voivat kantaa vastuuta ympäristöstä ja toimia esimerkkeinä organisaatiotason ilmastotoimissa.

Tämä diplomityö tutki koneoppimiseen perustuvan kysynnän ennakkoinnin ohjelmiston suunnittelua, käyttöönottoa ja käyttöä ylituotannon vähentämisessä osana LUT-yliopiston Waste Watch -projektia. Tutkimuskohteena oli Kampusravintolat Oy, LUT-yliopiston omistama ruokapalveluyritys. Tavoitteena oli arvioida dataohjatussa ennusteohjelmistossa vaikutuksia ruokahävikin määrään, CO₂-päästöihin ja ruokapalveluiden toimintaan.

Tutkimus toteutettiin toimintatutkimuksena, ja siihen sisältyivät hävikkidatan ja päästöjen analyysi, ohjelmiston toimivuuden arviointi sekä keskustelut henkilöstön kanssa. Pilotin aikana ennusteohjelmisto vähensi palveluhävikkiä keskimäärin 16 %, yksittäisinä kuukausina jopa 46 %. Tämä vastaa noin 30,5 tonnin vuosittaista CO₂-ekvivalenttipäästöjen vähenemää. Lisäksi työkalu tuki tuotannonsuunnittelua, kustannustehokkuutta ja henkilöstön sitoutumista ilmastotavoitteisiin.

Tulokset osoittavat, että kysynnän ennakkointi voi toimia skaalautuvana keinona ruokahävikin ehkäisyssä. Ennakoiva, dataohjattu ennustemalli vähentää ylituotantoa, pienentää päästöjä ja tukee ruokapalveluiden ilmastotyötä. Tämä diplomityö tarjoaa käytännön näkökulmia ruokapalvelutoimijoille, vastuullisuuspäälliköille ja tutkijoille, jotka kehittävät koneoppimispohjaisia ratkaisuja päästöjen vähentämiseen.

ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
CF	Carbon Footprint
LCA	Life Cycle Assessment
ML	Machine Learning
POS	Point of Sale
TWh	Terawatt hours

Table of contents

Abstract

(Abbreviations)

1	Introduction.....	3
2	Literature review.....	5
2.1	Food waste in institutional settings: challenges and solutions	5
2.2	Technology solutions in reducing food waste	8
2.3	Data-driven demand forecasting for food services	10
2.4	Review of previous initiatives to build demand forecasting for food services....	11
2.5	Summary of gaps the research study will address	14
3	Case study.....	16
3.1	Case context and pilot site background	16
3.2	Implementation of the project.....	17
3.3	Technical design and implementation	20
3.4	Overview of the demand forecasting tool.....	21
3.5	Forecasting model design and training process	22
4	Methods	25
4.1	Research methodology: Action Research	26
4.2	Stakeholder discussions	28
4.3	Practical understanding of food waste solutions.....	30
4.4	Estimating CO ₂ emissions.....	31
5	Results and findings.....	34
5.1	Quantitative reduction in food waste	34
5.2	CO ₂ emissions reduction.....	40
5.3	Operational impacts	42
5.4	Forecast accuracy, results verification and limitations	43
6	Discussion.....	46
7	Conclusions.....	50
	References.....	52

Appendices

Appendix 1. Benchmarking: Food waste technology solutions

Appendix 2. Summary of stakeholder meetings conducted for the thesis

Appendix 3. AI disclosure

Figures

Figure 1. Food flow and waste streams in campus restaurant operations.....	18
Figure 2. Timeline of Waste Watch project implementation.....	19
Figure 3. LUT Buffet's waste composition during Aug 2023 – July 2024	35
Figure 4. LUT Buffet's diner amount prediction example	36
Figure 5. LUT Buffet's menu item prediction example	37
Figure 6. LUT Buffet's service waste during Aug 2022 – May 2025	39
Figure 7. 20-day rolling median of relative forecasting error for Restaurant Skinnarila ...	43
Figure 8. 20-day rolling median of relative forecasting error for LUT Buffet.....	43

Tables

Table 1. Summary of digital food waste technologies in institutional settings	9
Table 2. Overview of food waste reduction and forecasting projects in Finland	13
Table 3. Steps of the action research cycle applied in Waste Watch project.....	27
Table 4. Thesis related seminars and workshops attended during the project.....	31
Table 5. Service waste reduction in kilograms	40
Table 6. Emission calculation	41

1 Introduction

In Finland, the food services sector including institutional kitchens, restaurants, and catering generates approximately 78 million kilograms of food waste annually and they represent around 13% of the total food loss and waste across the national food chain (Natural Resources Institute Finland LUKE, n.d.) Hereby, Food waste contributes to greenhouse gas emissions through the resources consumed in food production, transportation, storage, and preparation, while it also causes financial losses and unnecessary CO₂ emissions (Ritchie 2020.)

In campus restaurants with buffet-style dining, forecasting the number of daily diners is an everyday challenge and overproduction is common. Campus restaurants often lack the tools and systems necessary to make data-informed decisions about food preparation. Food production decisions are based on guessing, feeling, intuition and handwritten or otherwise collected historical data. However, predictive analytics, machine learning, and artificial intelligence-driven forecasting can help aligning food production with real-time demand. So far, the topic is underexplored in the context of Finnish university campus restaurants. This study investigates that research gap by exploring how data-driven systems can prevent overproduction and reduce the emissions in campus restaurants. This study is part of the Waste Watch project, funded by LUT University to help advance its climate efforts.

Recent years have seen increasing interest in predictive analytics, machine learning (ML) and artificial intelligence (AI) as tools for improving efficiency in various sectors, including food services. International software solutions such as Tenzo and Leanpath have succeeded in reducing waste in commercial catering environments. In Finland, digital tools like Hukka.AI, SmartKitchen and Biovaaka serve institutional kitchens, such as municipalities and schools. However, university campus restaurants operate under different conditions such as fixed lunch hours, self-service, unpredictable attendance, seasonality, international audience, and tight budgets. Published data on the impacts of demand forecasting in university campus restaurant settings is scarce. Food waste management software providers report promising reductions in waste and cost savings in different occasions, but there is a need for an independent research study. In addition, there is limited research on how machine learning-enabled demand forecasting in campus restaurants can be used to quantify and reduce CO₂ emissions.

Hereby, the aim of this study is to evaluate how data-driven, machine learning-enabled demand forecasting can support food waste reduction and CO₂ emission mitigation in campus restaurants. The study focuses on a real-life pilot project conducted in the Finnish university setting, specifically at Restaurant LUT Buffet and Restaurant Skinnarila operated by Kampusravintolat Oy on LUT University's Lappeenranta campus.

The following research questions guide the study:

RQ1. How can data-driven demand forecasting reduce food waste and associated CO₂ emissions in campus restaurants?

RQ2. What types of food waste and emission reductions can be achieved through forecasting?

RQ3. What are the key enablers and barriers to the successful implementation of forecasting tools in institutional food services?

For this study, Kampusravintolat Oy used a forecasting system developed by the Waste Watch project team to help plan food production more accurately with the help of machine learning. The study evaluates changes in food waste volumes and estimates CO₂ emissions reductions using quantitative methods. Data on meal production and waste volumes are analysed before and after the intervention, and the process of designing, creating and implementing the forecasting software is documented. CO₂ emissions and their reduction were estimated using the carbon footprint calculator embedded in the PoweResta menu management system, which provides average emission factors for various meal types based on standard life-cycle assessments. Using action research methodology, discussions with kitchen staff, managers and other stakeholders capture practical experiences, perceived benefits, and implementation challenges. As a result of the intervention, food waste and CO₂ emissions have been reduced, leading to cost savings and contributing towards institutional climate action.

This study is structured as follows: Chapter 2 provides a review of relevant literature, including food waste management practices, forecasting technologies, and recent projects in the field. Chapter 3 describes the case study and data collection process. Chapter 4 presents the research methods. Chapter 5 details the results of the study. Chapter 6 discusses the findings in the context of existing research, and Chapter 7 concludes with key takeaways and recommendations for future implementation and research.

2 Literature review

This chapter provides a literature review to contextualize and justify the research on data-driven demand forecasting to reduce food waste and CO₂ emissions in campus restaurants. The literature review follows a hybrid approach combining a systematic search and critical evaluation of academic and industry sources. The chapter begins by outlining the methodology and scope of the literature review, followed by the definition of key concepts and an overview of food service structures. The following sections then discuss current challenges and solutions in institutional food waste management, evaluates machine learning applications, and studies previous initiatives for demand forecasting. The final section identifies research gaps and explains how this study contributes to the field.

The hybrid literature review combined systematic search techniques with critical evaluation of the identified literature to ensure transparency and reproducibility of the review. A comprehensive search was conducted across multiple academic databases using LUT Primo, including access to ScienceDirect, Scopus and Web of Science. Search keywords included combinations of terms such as: 'predictive analytics', 'food waste forecasting', 'demand forecasting', 'machine learning in food services', 'artificial intelligence in food services', 'food waste in campus restaurants', 'food waste in food services', 'CO₂ emissions', and 'carbon footprinting'.

The articles included in literature review were published between 2014-2025, aiming for peer-reviewed articles when possible. Opinion pieces, blogs and publications from prior to 2014 were excluded. In addition to academic literature, insights from recent policy reports, project publications, and seminars were added to show recent examples and developments in the Finnish food service landscape.

2.1 Food waste in institutional settings: challenges and solutions

This section explores material flows, food waste generation, food waste management strategies and ongoing challenges in institutional food service environments. It discusses the importance of waste prevention and waste management regulations in Finland.

Food waste is defined as the reduction in the quantity or quality of food caused by the decisions and behaviours of retailers, food service providers, and end consumer (FAO 2025a). On the other hand, food loss is defined as food that discarded, incinerated, or

otherwise removed from the supply chain before reaching the retail stage, and which does not re-enter the system for other productive uses such as animal feed or industrial applications (FAO 2025b).

This study follows the categorization used in the Finnish food service sector, where food waste is divided into three main types:

- kitchen waste, which refers to food discarded during preparation,
- service waste, or food placed on the buffet line but not consumed, and
- plate waste, which consists of leftovers on customer plates.

(Finnish Hospitality Association MaRa 2021, 11)

Of these, service waste typically represents the largest share in institutional buffet settings and is the primary focus of this study.

The topic of food waste is critical to study now due to the urgent need to address climate change by reducing greenhouse gas emissions. According to Food Waste Index Report by United Nations Environment Programme (2024, 11), food waste is responsible for approximately 8-10% of global greenhouse gas emissions, covering both food loss and food waste stages. Additionally, the production of food that ultimately goes to waste occupies nearly 30% of the world's agricultural land (Ibid.). Annually, Finland generates approximately 351-376 million kilograms of food waste, with 17% originating from restaurants and catering services (Riipi et al., 2021). LUT University's Sustainability Report (2024, 28) indicates that food services account for 405.9 tons CO₂e in 2023, making it 16.8% of total carbon footprint (CF).

The article by Knorr and Augustin (2024) introduces a framework that places food waste generation within a broader human and systemic context, researching how social norms, cultural attitudes, and economic structures collectively contribute to food waste. This framework identifies four primary food waste generators:

- Human need, referring to basic physiological requirements and unavoidable losses in production
- Human intent, including aesthetic preferences and perceived food safety concerns
- Market constraints, such as pricing mechanisms and shelf-life limitations

- Food abundance, which includes the consistent availability of food throughout the year, reducing incentives for conservation (Knorr and Augustin 2024).

These factors are included within a wider system of external waste drivers, such as decisions to discard food for profit maximization, the pursuit of convenience, consumer confusion over labelling, and patterns of waste in household and food service settings (Knorr and Augustin, 2024).

Ma et al. (2023) studied the carbon footprint assessments of universities, particularly emphasizing Scope 3 emissions often representing the largest share of institutional carbon impact. Their review shows that data quality, methodological transparency, and tailored mitigation strategies are key to making university carbon accounting both actionable and credible (Ibid).

The purpose of reducing food waste through prevention is aligned with the waste hierarchy as established in the EU waste framework Directive 2008/98/EC. According to EUR-Lex (2024), the first step of the hierarchy is prevention, which involves measures taken before a material or product becomes waste, and the hierarchy is typically illustrated as an inverted pyramid, prioritizing the preferred options at the top and showing waste disposal to the bottom as the least favourable option. According to Silvennoinen (10.6.2024), under the waste hierarchy guidelines, a regulation mandates that starting from January 1, 2022, any unused but edible food must be redistributed for consumption wherever feasible but as of now, however, there are no penalties for non-compliance.

According to Kanwal et al. (2024), transforming edible food waste into new food products through upcycling is a promising pathway to valorise surplus and reduce environmental impact when prevention is not feasible. However, the authors stress that these strategies should complement, not replace, waste prevention efforts (Ibid). In line with the waste hierarchy, demand forecasting is preferable because it prevents overproduction from occurring in the first place (Ibid). The study also identified several barriers to effective implementation of circular food waste solutions, including limited staff training, regulatory ambiguity, and the lack of standardized procedures in professional kitchens (Ibid). These insights support the findings of this study, where the implementation of a data-driven forecasting tool was found to be effective but dependent on staff engagement, digital capacity, and organizational commitment.

Rodrigues et al. (2024) write that overproduction in food services is often a result of inaccurate forecasting leading to production of excess meals, while machine learning models can improve the precision of demand estimation and consequently reducing food waste. Also, earlier municipal studies show that kitchen staff often lack the tools or time to adjust portioning or production dynamically and food waste remains underreported unless digital monitoring is introduced (Hyvärinen, 2024).

2.2 Technology solutions in reducing food waste

This section will review common food waste management strategies and ongoing challenges in institutional food service environments.

Rodrigues et al. (2024) emphasize that overproduction in food services is often a result of inaccurate forecasting leading to production of excess meals, while machine learning models can improve the precision of demand estimation and consequently reducing food waste.

According to Hyvärinen (2024), several institutional food service providers report that a lack of real-time data, limited staff resources, and rigid production structures hinder proactive food waste prevention, and traditional practices often emphasize redistribution rather than avoiding waste avoidance. The same municipal study adds that the most impactful interventions are those that focus on preventing overproduction through flexible, data-supported planning systems (Hyvärinen, 2024).

As a practical example, Lukeloki is a digital monitoring tool developed by the Natural Resources Institute Finland (Luke) that helps food service providers to track and analyse different types of food waste in relation to food production and diner numbers, helping to reduce waste and improve cost-efficiency (Silvennoinen 2021.) During Soupster restaurant chain's Lukeloki waste tracking pilot in April 2024, 19 participating restaurants measured food waste daily, and the process revealed unexpected levels of kitchen and trimmings waste, inspired staff learning, and made it possible to calculate the monetary value of edible waste and its effect on profit margins (Soupster, 2024).

Another pilot study in collaboration with Sodexo and Food Waste Ecosystem reduced service waste by 25–50% and facilitated monthly donations of 800–1000 kg from schools and educational institutions in Helsinki and Vantaa (Sodexo, 2024). The initiative had positive feedback from staff and food aid partners, but the main challenges were equipment

failures, insufficient cooling capacity, and the unpredictability of diner numbers (Ibid). Overall, these findings support the needs for developing data-driven demand forecasting tools such as the one developed in Waste Watch project to enable more digitalised, automated and predictive food service planning.

This section provides an overview of existing software solutions and digital tools in the institutional food service sector. In Appendix 1 there is a big picture overview of the existing food waste technology solutions, and Table 1 lists food waste technology solutions used in institutional context. The digital solutions were benchmarked when preparing the software design for Waste Watch project.

Table 1. Summary of digital food waste technologies in institutional settings

Tool / Project	Approach	Method	Reported Outcome	Limitations
Leanpath	AI-enabled waste tracking tools, real-time analytics, behaviour change programs	Smart scales, devices with integrated cameras, dashboard	By reducing food waste by 50%, users can lower their overall food purchasing costs by approximately 2–6% (Leanpath, Inc. 2025.)	No forecasting, post-consumption action only
Biovaaka	Waste tracking	Smart scale + Dashboard	Users have typically succeeded in reducing food waste by 25–50% (Biovaaka n.d.)	Reactive only, limited integration with production planning
Hukka.AI	Waste tracking	Measuring, logging, and analysing different types of kitchen waste. Data visualization.	3–7% savings in raw material cost with 25-45% food waste reduction (Hukka AI Oy 2025)	No forecasting, focused on monitoring waste already generated
NextGen	Waste Forecasting + tracking	Diner forecast + staff-facing app	46% waste reduction in school meals (Alhonnoro, 2023b)	Focused on public education kitchens, limited emissions data

2.3 Data-driven demand forecasting for food services

This section includes an analysis of literature and industry insights related to demand forecasting in the context of food services.

Clark et al. (2024) conducted a systematic literature review identifying demand forecasting, waste reduction, and supply chain optimization as key application areas for AI technologies in food systems. The review found that machine learning models, particularly those based on time series, regression, and neural networks, can effectively predict consumption patterns and align food production with real-time demand (Ibid.) Such predictive capabilities are especially valuable in institutional settings, where overproduction is a major contributor to food waste (Ibid.) In addition, the study identified several barriers to successful implementation, including data quality issues, integration complexity, and limited organizational readiness (Ibid). These challenges align with the findings of these projects regarding the implementation of forecasting tools in campus kitchens.

Machine learning applications in the restaurant sector include demand forecasting, estimating visitor numbers, analysing customer reviews, and improving revenue strategies (Chatterjee 2022, 127). For example, Hübner et al. (2024) concluded that bakery-specific machine learning-based demand forecasting systems can reduce food waste by up to 30% and found it beneficial to integrate external factors such as weather and holidays into the model. Additionally, machine learning can be applied to various aspects of food waste management beyond demand forecasting such as optimizing food treatment processes, composting, biogas production, and landfill monitoring, improving sustainability across the food system (Chatterjee 2022, 126-127).

The study by Rodrigues et al. (2024) compared multiple machine learning models with baseline models, simulating real-world catering environments and provided quantified performance improvements across multiple canteens. Causal models like Light Gradient Boosting Machine include contextual variables such as menu types, weather, and holidays, which enhance predictive power in environments with short horizons (Ibid). Time series models such as Long Short-Term Memory (LSTM) and Transformer showed potential in larger datasets, with LSTM outperforming in high-volume settings (Ibid). Overall, LSTM models yielded the best results in a large company canteen, cutting food waste by over 50% (Ibid).

Schwartz et al. (2019) show exponential growth in computational demands of AI, with model training requirements increasing by 300'000 times between 2012 and 2018. Currently according to the International Energy Agency (IEA, 2025), the global data centres used 415 TWh, or 1.5% of global electricity by 2024 and by 2030, total data centre electricity consumption is expected to more than double to 945 TWh, making it larger than Japan's current national demand. Building on this, Menzies et al. (2024) introduced the concept of Green Software Engineering, which includes sustainability considerations throughout the entire software lifecycle. Reflecting further on sustainability of machine learning solutions, Hübner et al. (2024), did life cycle assessment (LCA) to evaluate both the direct environmental impact of using an AI tool for demand forecasting and the environmental benefits from reducing food waste, concluding that reducing food waste with machine learning had a larger environmental benefit than the emissions caused by running the machine learning forecasting system.

2.4 Review of previous initiatives to build demand forecasting for food services

In recent years, several national and international initiatives have addressed the issue of food waste in institutional food service environments with a growing emphasis on predictive technologies, data utilization, and behavioural change. This section reviews selected projects relevant to this study, particularly those that have introduced or tested demand forecasting, smart kitchen systems, or systemic approaches to reducing waste in large-scale meal production settings.

The material presented here is based primarily on the expert discussions and project overviews shared during the Hävikkihankkeiden Pyöreä Pöytä seminar, held in Helsinki on 13 March 2025, organized by OSAO and Luonnonvarakeskus (Luonnonvarakeskus & OSAO 2025).

Several projects in Finland focus on integrating technology solutions into institutional food services and are listed in Table 2. For example, the DiningFlow project by University of Turku has developed data-driven solutions using sensors and kitchen system integration to forecast demand in self-service restaurants (Luonnonvarakeskus & OSAO 2025).

In the field of forecasting solutions, the NextGen Forecasting Model 3.0 developed by Waste Insight team at the University of Vaasa makes daily and weekly predictions of total waste, plate waste, and kitchen waste (Kuusisto, 2025). In addition to the technology solution,

NextGen project uses socio-behavioural change strategies using participatory methods such as ideation workshops, nudging tools, model of continuous improvement, and gamified task cards to involve kitchen staff in daily waste reduction practices (Ibid).

Besides that, Projects such as Ce4Re (Circular Economy for Restaurants) led by Satakunta University of Applied Sciences and various EU partners, offer broader circular economy strategies for the restaurant and food sectors contributing to the sustainability transition on industry level.

On the other hand, behavioural and socio-technical dimensions are still underexplored in this field, but Nahistuneet naatit project by Lapua City, University of Vaasa and University of the Arts Helsinki uses artistic engagement to influence student behaviour in school cafeterias.

Additionally, HävikkiHub project by SeAMK, XAMK and Finnish Professional Catering Association focuses on training and knowledge-sharing among food service professionals. Together, these projects form a growing ecosystem around the use of data, technology, research, and collaboration to make food services more sustainable (Luonnonvarakeskus & OSAO 2025).

Table 2. Overview of food waste reduction and forecasting projects in Finland

Project	Consortium	Core Focus	Timeline
Älykäs ruokahävikki haltuun	OSAO, OPAO & Luke	Reducing food waste in vocational school food services through smart solutions and education	08/2024 – 07/2027
Nahistuneet naatit	University of Vaasa, Lapua city & University of the Arts Helsinki	Reducing plate waste in schools using art-based engagement and student participation	08/2024 – 07/2026
Ce4Re – Circular Economy for Restaurants	SAMK + EU partners	Developing circular economy solutions for restaurants in Central Baltic region	09/2023 – 06/2026
DiningFlow	University of Turku & industry partners	Using smart technology and forecasting tools to reduce waste in self-service restaurants	2023 – 2025
Food Waste Ecosystem	Vantaa, Helsinki, Luke & Laurea	Building a food service ecosystem to reduce food waste in the Helsinki region	04/2023 – 03/2026
HävikkiHub	SeAMK, XAMK and Finnish Professional Catering Association	Training and tools for professional kitchens to manage and reduce food waste	10/2024 – 03/2026
LowInFood	University of Tampere + EU-partners	Piloting innovative solutions for low-waste food value chains across Europe	Late 2020 – Early 2025
NextGen	University of Vaasa	Developing forecasting models and behaviour change tools for buffet dining	04/2023 – 03/2025
StratKit+	University of Helsinki (Ruralia Institute) + EU partners	Expanding sustainable public meal toolkit with focus on procurement and food waste	10/2022 – 03/2024
Urban & Local	Forum Virium Helsinki + collaboration partners	Creating a sustainable urban food ecosystem and piloting smart food solutions	01/2023 – 12/2025
WiseFood	City of Turku	Piloting circular economy innovations for local food systems in Turku	09/2023 – 11/2025

2.5 Summary of gaps the research study will address

While a growing interest towards reducing institutional food waste through data-driven approaches came up in the literature review, several research gaps were identified particularly in the context of campus restaurants and their unique operational environments. This study aims to address the following limitations identified in existing research:

1. Limited focus on campus restaurants:

Although food waste in institutional settings has been widely studied, there is a lack of targeted research on campus restaurants. Campus restaurants operate with specific constraints such as fluctuating student numbers, governmental subsidies and academic schedules.

2. Integration of real-time data:

Many existing forecasting or waste tracking systems rely on static inputs or manual logging. While some recent studies emphasize the value of contextual inputs such as menu type, weather, and holidays for improving forecasting precision (Rodrigues et al., 2024), the integration of real-time data streams remains limited in practical implementations (Clark et al., 2024). These limitations affect the scalability and usability of such tools in institutional kitchens but could be solved using API (Application Programming Interface) solutions.

3. Lack of combined food waste and CO₂ analysis:

Very few forecasting models simultaneously assess the environmental impact of food waste. While the importance of Scope 3 emissions is recognized in higher education institutions (Ma et al., 2023), most solutions still lack integrated carbon footprinting or life cycle assessment (LCA) approaches.

4. Decision-making support for end users:

Food waste solutions tend to focus on retrospective reporting rather than offering predictive or real-time feedback to kitchen staff. Tools such as Biovaaka or Leanpath are brilliant in tracking existing waste but have limited decision support for adjusting production in advance or finding the crucial information for decision-making among all the data.

5. Forecasting output is not optimized for operational use:

Some existing tools forecast in kilograms, sales numbers or diner numbers, which may not align with campus kitchen practices that operate using container and tray-based calculations for cooking amounts. This mismatch can reduce the usefulness of the forecasting output in everyday decision-making

By responding to these research gaps, this study contributes to the emerging field of sustainable food service innovation by piloting a machine learning enabled forecasting system that integrates real-time POS and menu data, provides container-level predictions, and enables CO₂ impact estimation. The findings of the study aim to support both climate action and operational efficiency in campus restaurants.

3 Case study

This chapter describes the implementation process of the Waste Watch project from June 2024 to May 2025. It provides a detailed description of the case study setting, technical system deployment, and stakeholder collaboration. It begins by describing the institutional context of LUT University's food services and the reasons for selecting the pilot site. Following sections explain the integration of the forecasting tool, covering the steps taken for data acquisition, implementation for operational use, as well as technical and organizational challenges encountered along the way.

This chapter contributes primarily to Research Question 3 by identifying key enablers and barriers to successful forecasting tool implementation in institutional food services. It also provides insights into Research Question 1, by analysing how data infrastructure and staff engagement affected the forecasting system's impact to reduce overproduction. Elements of Research Question 2 is also touched upon, as the chapter documents how improved planning practices began to influence surplus, waste and emissions generation patterns during the pilot phase.

3.1 Case context and pilot site background

Initially, the project team was interested in collaborating with both campus restaurants in Lahti and Lappeenranta. On May 23rd 2024, the Waste Watch project team did a site visit at LUT University's Lahti campus restaurant to discuss potential project collaboration with restaurant manager. The meeting gave insights into the Lahti campus restaurant's daily operations, food waste monitoring practices, and use of technology solutions. However, after the Lahti restaurant decided not to participate in the project, cooperation was initiated with Kampusravintolat Oy in Lappeenranta. Kampusravintolat Oy was founded in 2015, is co-owned by LUT University and LAB University of Applied Sciences and operates three restaurants in Lappeenranta: Restaurant Skinnarila on the LAB campus, and LUT Buffet and Street Cafe on the LUT campus (Kampusravintolat Oy, 2025).

Kampusravintolat Oy, starting with LUT Buffet and later extending to Restaurant Skinnarila, was selected as the pilot site due to its centralized data infrastructure, experience with digital monitoring tools, availability of historical data, and openness to innovation. The

collaboration was supported by key personnel, including the restaurant manager and head chef.

The first project planning meeting took place on 12th of June 2024, and the purpose was to assess the feasibility of integrating a predictive analytics solution into the restaurant's existing operations and digital systems. Key discussion areas included current waste measurement practices, the use of Biovaaka smart scales, the SKJ point-of-sale system, and the PoweResta kitchen management software. Possibilities for API (Application Programming Interface) integration between these systems and the new forecasting tool were also explored.

According to Selpy et al. (2019), an API specification defines the available services of a software application, detailing the required inputs, the structure of the outputs, and the protocol used typically over the web for automated data exchange between computer programs. Considering the usefulness and efficiency of API functionality over sending data manually, access to data from PoweResta and SKJ was identified as essential to ensure accurate forecasting.

Building the demand forecasting software would also require access to historical data that would be used for training the forecasting model. In terms of data availability, data from 2020–2021 had to be excluded due to the unusual circumstances caused by the COVID-19 pandemic. As a result, only data from 2022 to 2024 was used to train the machine learning model.

Additional insights were shared about operational forecasting challenges, such as daily diner numbers varying by up to 500 people, as well as the potential for future nudging interventions to influence customer behaviour at the buffet. While nudging strategies were deemed outside the scope of the current project, they could be part of future research (Mattila, 2024d.)

3.2 Implementation of the project

From the point of view of the project goals, research design, and software design it was important to understand what happens for the food and waste streams in campus restaurants. Figure 1 illustrates the food flow and waste streams within the restaurant operations at Kampusravintolat Oy restaurant LUT Buffet as observed during the Waste Watch project.

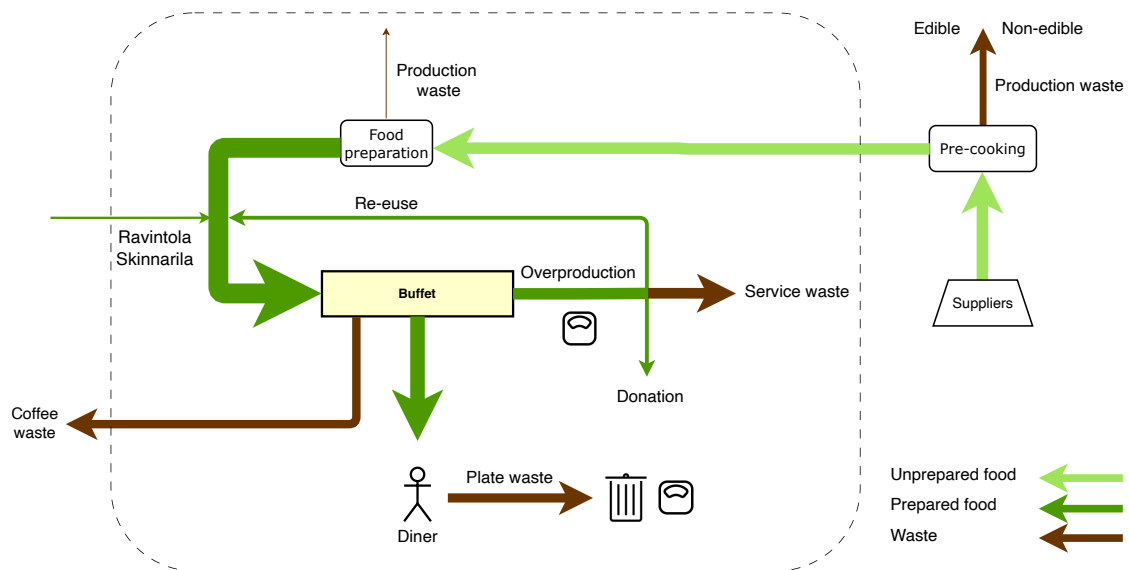


Figure 1. Food flow and waste streams in campus restaurant operations

1. The restaurant operations process begins with raw ingredients delivered by suppliers, which enter the pre-cooking phase. At this stage, production waste is generated and categorized into edible and non-edible fractions.
2. Following pre-cooking, food goes to the preparation area where additional production waste may occur. Prepared food is then transferred to the buffet, which is a central point of food distribution to diners.
3. Overproduction at the buffet results in food waste because food was prepared but not consumed, but which can be either reused, sold at a reduced price, or donated.
4. Diners contribute to plate waste if edible food is discarded after serving, and specific waste streams such as coffee waste are separately managed.

The figure shows three types of flows: unprepared food (light green), prepared food (dark green), and waste (brown), visually emphasizing critical points where food is lost or recovered within the system. By mapping these flows, the diagram helps to understand how overproduction, reuse, donation, and disposal occur in institutional food service operations.

Figure 2 shows the timeline of the project implementation and main milestones.

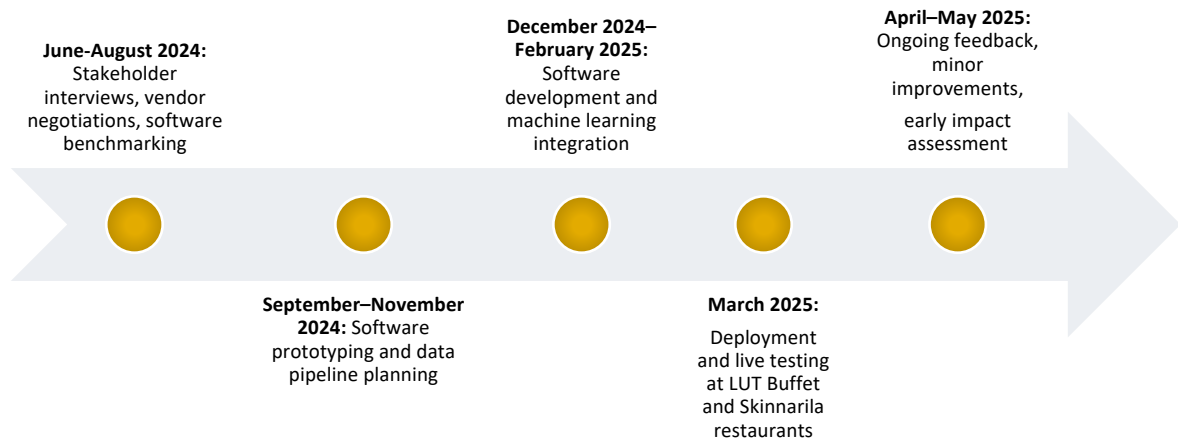


Figure 2. Timeline of Waste Watch project implementation

The Waste Watch team chose to build a demand forecasting tool using historical sales and production data, aiming to better match food preparation to actual diner numbers. This data-driven approach allowed real-time adjustments, increased planning accuracy, and ultimately reduced waste volumes and greenhouse gas emissions.

The Waste Watch project involved a small interdisciplinary team. The project team primarily consisted of junior researcher Jubeen Sharbaf and the thesis author Sara Mattila. At the initial stage, the team also received advisory support from Juha Vänskä from the University of Vaasa. Jubeen Sharbaf led the research design, developed the forecasting software, collaborated closely with the restaurant staff, and coordinated the overall project. Sara Mattila focused on literature review, data collection, communication with stakeholders and vendors, analysed waste and emissions data, as well as synthesized findings from discussions with stakeholders. In addition, a freelance software developer set up the technical infrastructure and implemented the machine learning components. Also, a student assistant was hired on two occasions to support data collection and another student contributed by writing a separate thesis focusing on plate waste. Besides that, the project involved over 20 meetings with stakeholders, which are listed in Appendix 2.

The Waste Watch project encountered several challenges during its implementation, but these were met with practical solutions. One issue was that the point-of-sale (POS) vendor

took nearly four months to establish API access to daily sales data. Daily sales data is a crucial element of the software, and during the delay it had to be sent manually until establishing the API connection.

The delay created pressure on the project timeline. Fortunately, the team was able to progress quickly after getting all the API connections eventually and managed to deliver a functional software within the project timeline. Also, most of the API connections were obtained for free or with just an installation cost and small monthly fee, avoiding the high integration and monthly costs that had occurred in other food waste related forecasting projects.

Another challenge came from unexpected overhead costs. This meant that some initially budgeted services, like hiring an external consultancy to verify CO₂ emission reduction calculations, had to be cancelled. To address this, the Waste Watch team created CO₂ emissions dashboard into the software using internal resources.

3.3 Technical design and implementation

Before developing the demand forecasting tool further, the Waste Watch project team had several meetings and discussions to learn from stakeholders involved in similar food waste prediction initiatives. The goal was to gather practical, experience-based insights that are not typically available in academic literature.

The first meeting with project advisor with extensive experience on predictive models and food waste forecasting through University of Vaasa's NextGen and Wasteless projects took place on 23rd of May 2024. In that meeting it was advised that predictive software solutions typically require 3–4 months of data to train their predictive model and hereby the importance of starting data collection early and maintaining consistent data quality was taken into consideration (Mattila 2024g). Integrating key datasets was considered important, and specifically this meant accessing menus and recipes via a menu management system, historical and actual sales data through point-of-sale (POS) systems, and waste measurement data from smart scales (Ibid). It was also recommended incorporating variables such as weather, holidays, and student absences through either external platform integration or direct collaboration with the institution to enhance forecast accuracy (Ibid). These recommendations were particularly useful for the project and became one of the core foundations for designing the Waste Watch software architecture.

Another benchmarking meeting was held with developers and project managers from the University of Vaasa's NextGen and Wasteless projects on 27th of August 2024. Their systems used a Python-based linear regression model with a user interface deployed on Microsoft Azure, relying on AJAX calls to retrieve real-time data from connected platforms (Mattila, 2024e). Also, in that discussion it came up that connecting APIs might be expensive or technically demanding, and that it could be more feasible to begin with manual data entry (Ibid). However, in the Waste Watch project, API access was obtained relatively quickly and for free from most of the vendors. Only one installation fee and a modest monthly subscription were required.

During the early development planning, a lecture on software architecture and procurement at Helsinki Incubators by University of Helsinki gave practical advice for early-stage software projects, mentioning that many software projects fail because they are planned like fixed products instead of being developed step by step (Mattila, 2024f). Learning from this, the Waste Watch project used an agile approach with short development rounds, where regular feedback from chefs, vendors, and restaurant management actively shaped the tool's development in iterative cycles according to the principles of action research.

Besides that, the lecture at Helsinki Incubators by University of Helsinki mentioned that when developing a tool where reliability and long-term use matter, involving experienced developers early on can significantly improve the quality and maintainability of the software (Mattila, 2024f). This guided the Waste Watch project team to contract a senior software developer specialized in machine learning to build a working proof of concept including core features. Next section will give an overview of the demand forecasting tool.

3.4 Overview of the demand forecasting tool

According to Sharbaf (2025), the predictive forecasting tool Waste Watch uses three years of historical sales data and live daily sales data from the LUT Buffet restaurant to generate machine learning-based predictions. A key innovation of the system is that predictions are provided in units familiar to kitchen staff such as number of containers, therefore supporting easy daily decision-making (Ibid.) The development process began by standardizing data entry formats for chefs and creating structured input forms that enable continuous measurement of production quantities (Ibid).

The software solution integrates data from multiple sources to predict demand:

- Point-of-sale system (SKJ by Confirma Oy)
- Menu management platform (PoweResta Oy)
- Smart scale data (Biovaaka Oy)

(Sharbaf 2025).

One of the software's key strengths is that besides demand predictions, it connects to environmental and cost impact estimates. The following sections describe the system's forecasting model and training process.

3.5 Forecasting model design and training process

Kananen and Puolitaival (2019, 45-46) explain that machine learning algorithms are classified based on how data is provided: supervised, unsupervised, and reinforcement learning. A machine learning model is created by training it on data to identify patterns as well as dependencies, and once trained, the model can make predictions on new data by generalizing from the training set (Ibid). AI systems require large amounts of data to learn, which can come from various sources such as customer interactions, sales, websites, sensors, and documents (Ibid.) This data may be structured or unstructured such as text, images, sensor outputs, or financial records (Ibid.) Preparing the data is a crucial and often time-consuming step as data formats vary, but the goal is to create a clean, consistent, and coherent dataset because poorly prepared data can lead to poor model performance (Ibid.)

The model used for daily diner count prediction at the LUT Buffet is a machine learning model developed using Google ML. It forecasts customer volume up to several weeks in advance, providing actionable data for production planning, as production plans are done few weeks before production date. Training was performed on a dataset spanning three academic years.

Implementing data-driven forecasting in campus food services required systematic data collection and data quality monitoring. The model incorporates a set of contextual features selected to reflect both temporal patterns and operational factors, including:

- Historical daily sales volumes
- Seasonality and weekly fluctuations
- Academic and national holiday calendars
- Days since the start of the academic year

- Planned daily menu (converted into model-compliant input formats)
- Weather data

Menu-related inputs were pre-processed into structured formats compatible with the model's requirements. Forecast outputs were translated into container units, making them directly usable by kitchen staff and supporting integration with production planning tools. Although waste measurement data was not used as a direct input, it was used for evaluating downstream outcomes such as overproduction and environmental impacts.

The forecasting model is deployed and operated entirely online, enabling cloud-native training, inference, and querying. Predictions are generated via structured SQL calls embedded in the application's backend. Planned improvements include preserving historical forecast versions for longitudinal analysis and studying how predictive accuracy influences actual kitchen decisions and food waste outcomes.

GastroNorm (GN) containers are standardized food containers codified in the European standard EN 631, and their uniform sizing allows for efficient workflow, space-saving combinations, as well as simplified planning in institutional kitchens (New Gastroline GmbH. 2025). In the context of this project, forecasting food quantities in GN containers, named in the software as trays ('pakki' in Finnish), rather than kilograms was essential to align the predictive model with the kitchen's actual production and service routines.

As part of data integration process, the Waste Watch development team introduced a new system to associate each dish in the forecasting model with its corresponding tray type and average tray weight. This allows for more accurate estimation of consumption per diner and is useful in visualizing the environmental and economic impact of demand forecasting. A new internal dashboard page called "Trays and Unit Cost" was created to manage this data. The matching logic is based on two key variables: "Main Ingredient" and "Dish Type". These two fields are manually updated by the team on a weekly basis, ensuring that each menu item in the Food options dataset can be connected to a valid tray type using predefined combinations. Once a valid tray type is assigned, the average tray weight is filled using values collected and validated from real-world measurements. These weights are gradually refined through repeated observations and stored in a .csv file.

To reduce human error and improve consistency, the system restricts allowable combinations of Main Ingredient, Dish Type, and Tray Type based on known mappings.

However, some mismatches or missing data have occurred. For instance, recipe classification errors such as Asian-style barley-vegetarian dish being switched from classification of stew to special dish, which temporarily broke the tray-type matching logic. Such issues were resolved by manually reviewing and updating entries in the database.

Once the data was synchronized across the relevant tables, the developer ran a batch calculation of historical consumption per diner, based on tray weights and sales volume. This step enabled estimation of how much food (by weight) was prepared and served per person on average.

These values formed the basis for quantifying:

- Per-dish and per-day food waste
- CO₂ emissions per serving
- Potential cost savings from reduced overproduction

In future, this tray-type and unit-cost page is intended to be part of the user-facing dashboard. Since the unit price values were extracted from pre-pandemic BioVaaka data, they will require periodic updates by the restaurant to maintain financial accuracy. Ultimately, this setup supports the project's goal of showing partner restaurants how much they save by reducing food waste, emissions, and costs by using the forecasting system. Next chapter will discuss the methods and materials used in the research process.

4 Methods

This chapter describes the research methodology used to investigate the impacts of data-driven demand forecasting on food waste and CO₂ emissions in campus restaurants. The research was conducted as part of a pilot project; therefore, it is essential to differentiate between the research methods used to evaluate the pilot, and the project implementation steps taken to develop and deploy the forecasting tool.

The research methodology refers to the action research approach guiding the study, including the data collection techniques (e.g., stakeholder discussions, forecasting data, waste measurement), data analysis, and evaluation frameworks used to answer the research questions. In contrast, the project implementation sections describe the technical development and deployment of the forecasting system in a real-world context. These sections are included to provide transparency on how the pilot study was constructed, as this context is necessary for understanding the empirical results. However, the implementation details are not themselves the focus of scientific analysis. Instead, they rather provide the practical foundation upon which the research is built and evaluated.

The thesis research was commissioned as a part of LUT University's internal climate enhancement project called Waste Watch occurring from June 2024 to May 2025. The research questions were developed by combining insights from the academic literature on food waste reduction and data-driven forecasting with the practical objectives of the Waste Watch project at LUT University. A review of literature and earlier case studies indicated that predictive technologies could reduce food waste and create measurable environmental and financial benefits. For example, according to Alhonnoro (2023a), University of Vaasa tested predictive software in a school in Lapua, where it reduced food waste by 46%, cutting average waste per diner from 90g to 48g and saving around 2000 euros per month. These findings helped forming the research questions that would explore both the impact of demand forecasting tools, and the conditions required for their successful adoption in food services context.

The chosen research questions aim to address:

1. The potential of data-driven forecasting to reduce food waste and greenhouse gas emissions,

2. The types of reductions in food waste and emissions achievable through forecasting, and
3. The factors that either support or hinder effective implementation.

The questions are designed to produce actionable knowledge that supports both academic research and real-world sustainability impact

4.1 Research methodology: Action Research

Action research progresses through a cycle of preparation, implementation, monitoring, and reflection (O'Brien, 2001). This method typically involves collecting data prior to an intervention to establish a baseline, introducing a solution in practice, and then gathering follow-up data to evaluate its effects and inform further development (Ronen, 2019). According to McNiff (2013, 25), action research typically consists of four key phases: action, research, reflection and evidence, and knowledge creation. The action phase involves engaging directly with the context under study, examining current practices, organizational conditions, and underlying assumptions, while being mindful of broader institutional, social, and political dynamics (Ibid).

In this study, the action phase was conducted by doing kitchen visits, stakeholder discussions, literature review and review of previous initiatives for food waste reduction. The research phase consists of systematically gathering data related to those actions and their environment (McNiff 2013, 25). In this study, this was achieved through stakeholder discussions, food waste tracking, monitoring and analysing data regarding menus, sales, costs and emissions, and feedback collected during the pilot deployment. The reflection and evidence phase are focused on analysing the data to understand the impact and effectiveness of the actions, forming a basis for future improvement (McNiff 2013, 25). In this project, reflective practices guided ongoing software adjustments and stakeholder communication. Finally, knowledge creation refers to generating insights and drawing conclusions that can inform future practice or research (McNiff 2013, 25). The conclusions from this process contributed to both operational improvements in the restaurant and academic knowledge on digital forecasting tools for sustainability. This iterative approach was ideal for integrating a data-driven solution in a dynamic food service environment where production decisions and customer behaviour change daily, and where stakeholder engagement is needed for practical implementation.

The method was applied through a series of structured, real-world interventions and evaluations. Table 3 lists the stages of the action research cycle and how each phase was implemented during the thesis project.

Table 3. Steps of the action research cycle applied in Waste Watch project, adapted from O'Brien (2001), McNiff (2013) and Ronen (2019)

Step	Description	Timeline
Identify the problem	Food services were identified as a major contributor (16.8%) to LUT University's carbon footprint (LUT University 2014, 28).	March 2024
Plan the intervention	- Baseline data (food waste, diner counts, menus, emissions) were collected to understand the current situation. - A data-driven forecasting software was designed. - Freelance software developer was contracted. - Software development began based on the data analysis and identified needs.	June 2024 to October 2024 Software development started in November 2024.
Implement the solution	- The forecasting software was piloted in campus restaurant operations. - Kitchen staff were trained, and adjustments were made during deployment.	Software in use from March 2025.
Collect data	Data were gathered before and after the intervention: biowaste volumes, customer numbers, menu data, and estimated CO₂ emissions.	June 2024 to June 2025
Reflect on outcomes	Experiences of kitchen staff and operational feedback were gathered through discussions, data monitoring and informal discussions. Adjustments were made based on staff input.	March to June 2025
Analyse results	- Quantitative results were assessed. - CO₂ savings, cost savings, waste reduction, and staff insights were used to evaluate impact.	April to June 2025

To summarize, structured action research helped the Waste Watch project team to design, co-develop, test, and iterate the demand forecasting solution in close collaboration with the end users, kitchen staff and managers at Kampusravintolat Oy. The frequent collaboration and discussion with the end users also kept the research connected to the day-to-day

activities at campus restaurants. Consequently, the study resulted in practical improvements for institutional sustainability practices, which was one of the goals of Waste Watch project.

4.2 Stakeholder discussions

As a part of the project, many stakeholders and specialist discussions were conducted to gather information via unstructured interviews before designing the software, while it was being developed, and after implementing it. The conversations were deliberately unstructured, and the facilitator brought an indicative topic list covering discussion areas. The discussions were not recorded but the Waste Watch project team members took notes from the meetings and expanded them into detailed memos within two hours of each session. Limitations of these discussions are that note-based unstructured discussions lack the depth of recorded, transcribed interviews. Immediate memo-writing and going through the notes together with the project research team partially mitigate this.

Before designing the software and while it was being developed, purpose of the unstructured stakeholder discussions was to obtain insights regarding food services operations, food waste reduction practices, and existing food waste-related software and technology solutions. After implementing the demand forecasting software at Kampusravintolat Oy, several discussions were conducted to understand end user's experiences of the software, and any challenges or development opportunities that could be addressed.

One key discussion was conducted on 6th of February 2025 at a Kampusravintolat kitchen on LUT University's Lappeenranta campus. The discussion included two staff members: the restaurant manager, and the head chef. The purpose of the discussion was to understand how daily food production volumes are currently estimated and how forecasting data could potentially support decision-making (Mattila and Sharbaf, 2025).

The unstructured interview revealed that kitchen staff primarily rely on experience and so-called "chef's secret notebooks" to estimate daily tray quantities for production. Although preliminary forecasts for the number of diners are available, they are not yet used in a systematic or real-time way to guide preparation decisions. Instead, staff assess food availability during service and adjust production volumes reactively (Mattila and Sharbaf, 2025).

The kitchen team practices partial preparation as a waste prevention strategy, particularly for expensive ingredients such as meat. In dishes like casseroles and lasagna, inexpensive base elements such as sauces are prepared in the morning, while more costly protein components are added only as needed. However, surplus base elements are often discarded if unused, revealing an area where improved forecasting could reduce waste (Mattila & Sharbaf, 2025). The discussion also highlighted important contextual factors affecting demand, such as customer food preferences (e.g. the popularity of chicken), daily menu combinations, and variation in tray sizes. Staff emphasized that any forecasting system must reflect this complexity and fit naturally into their workflow. Additionally, they noted that prediction tools showing required trays per dish could be particularly helpful for less experienced chefs (Mattila & Sharbaf, 2025).

The stakeholder discussions with the kitchen staff brought a deep understanding of tray-based production planning, partial preparation practices, and situational decision-making were crucial for developing the software solution. These findings helped steer the research towards a more practical approach regarding usability, relevance for daily operations, and potential for actual food waste and CO₂ reductions. Action research has several advantages, particularly in applied research contexts that need practical problem-solving and continuous stakeholder collaboration. As O'Brien (2001) states, a key benefit of action research approach is its cyclical and participatory nature, which allows researchers to engage in reflective cycles of action and learning. This adaptability was needed for real-time improvement and made the method well-suited for improving sustainability practices at institutional food services. In the context of this study, action research included close collaboration with kitchen staff, iterative testing of a data-driven forecasting solution, and the practical integration of feedback into both system development and daily operations.

Several improvements, such as modifications to food production routines, changes in planning practices, and improved understanding of forecast interpretation were implemented during the pilot phase. These practical changes were brought up thanks to the participatory structure of action research. They also contributed to a shared sense of ownership among stakeholders and made the project more relevance to operational decision-making. O'Brien (2001) further notes that action research often leads to results that can be immediately applied, which proved to be the case in this project. Real-time insights from restaurant

management, kitchen staff and researchers were used to refine the tool's usability and adapt it to fit with kitchen workflows.

However, several limitations were also encountered. As O'Brien (2001) points out, action research can be time-intensive and resource demanding, particularly due to the need for repeated stakeholder engagement, feedback cycles, and ongoing adaptation. This was indeed the case in the Waste Watch project where significant amount of time was used for unstructured stakeholder discussions, presentations, training kitchen staff, restaurant visits, collaborative workshops, taking measurements, analysing feedback and data, and technical iteration of the software. Another limitation lies in the contextual nature of the findings. Since the study was conducted in a specific university food service setting, its results may not be directly generalizable without modification.

Furthermore, O'Brien (2001) warns that researchers must remain aware of their own biases when acting as both facilitators and observers. To address this, the project team had reflective documentation practices where notes from each project-related meeting and activity were shared and discussed among the project researchers to include different viewpoints and avoid bias. In summary, while the action research approach required a considerable investment in time and coordination, it facilitated practical outcomes, deep understanding of the operational challenges, and stakeholder-driven innovation.

4.3 Practical understanding of food waste solutions

In addition to the action research formal stages of planning, acting, observing, and reflecting (O'Brien, 2001), the researcher participated in food waste related events, site visits, and professional seminars. These events were not part of data collection but were intended to deepen the practical understanding of current food waste solutions, stakeholder roles, and system-level challenges. Engaging with industry professionals, municipal actors, and fellow researchers in a variety of events gave practical insights into challenges and opportunities in food waste related logistics, policy, digital tools, and food waste valorisation. Insights gathered in these events were used to enrich the thesis with recent, practical examples of food waste reduction in institutional settings. The information gathered in the sector-specific events also helped the development of the demand forecasting tool through understanding other food waste related digital solutions in the market. A summary of the key events and their relevance to the thesis is presented in table 4.

Table 4. Thesis related seminars and workshops attended during the project

Date	Organizer	Core Focus	Key learnings
8.2.24	Food Waste Ecosystem, LUKE	Best practices in food waste reduction	Learned about University of Vaasa's NextGen project and predictive modelling, which served as inspiration for Waste Watch project
18.4.24	Food Waste Ecosystem	Best practices in donating surplus food	Regulation, practices and actors in food donations and food aid
30.5.24	Food Waste Ecosystem	Food waste valorisation and utilization	Opportunities and challenges valorising and utilizing edible and non-edible food waste
4.9.24	Helsinki Incubators, University of Helsinki	Lecture on software architecture and procurement	Learning the basics of software architecture and procurement for project purposes
27.9.24	Food Waste Ecosystem	Hävikkiiviikko Seminar	Various seminar presentations and panel discussions on food waste reduction
22.11.24	LUT University	Visit to Ämmässuo Eco-Industrial Centre and Vantaan Energia Waste-to-Power plant	Understanding what happens to food waste in waste management facility
12.12.24	Laurea Kiertotalous Living Lab	Business from food waste side streams	Business opportunities regarding food waste side streams
13.3.25	LUKE & OSAO	Meeting of food waste related research project teams in Finland	Overview of all the food waste related projects in Finland
22.5.25	Food Waste Ecosystem, City of Vantaa	Food waste and food surplus logistics and robotization	Challenges in food waste and food aid logistics, opportunities with robotization of food logistics
27.5.25	Biovaaka, LUT & University of Vaasa	Project conclusions and software updates	Results of Next Gen and Waste Watch projects

4.4 Estimating CO₂ emissions

When evaluating the environmental impact of products or services, two commonly used methods are Life-Cycle Assessment (LCA) and its subcategory Carbon Footprinting (Jolliet et al. 2015). Both support environmental decision-making but are different in their scope,

purpose, and the kind of results they produce (Sahota 2014, 41–42). Life-Cycle Assessment is a method used to measure the environmental effects of a product or service throughout its entire lifecycle from raw material extraction to use and disposal. This approach includes four key stages: goal and scope definition, life cycle inventory, impact assessment, and interpretation (Sahota 2014, 40–41). Jolliet et al. (2015, 15–16) state that LCA looks at many different environmental impacts, not just climate change and it helps to compare how different choices affect things like pollution, water use, or resource depletion, and not just carbon emissions while Carbon Footprinting is a more targeted approach that focuses only on greenhouse gas (GHG) emissions, typically expressed as carbon dioxide equivalents (CO₂e). Carbon accounting is based on five main principles of relevance, completeness, consistency, transparency, and accuracy (WBCSD & WRI, n.d., 7). In the LUT Waste Watch project, carbon accounting and carbon footprinting was chosen over a full Life-Cycle Assessment (LCA) due to the intended use of the data for climate impact reporting and climate target setting.

According to the Greenhouse Gas Protocol, emissions are divided into three scopes:

- Scope 1: Direct emissions from owned or controlled sources,
- Scope 2: Indirect emissions from the generation of purchased energy, and
- Scope 3: Other indirect emissions from sources not owned or controlled, such as the full value chain

(Sahota 2014, 39–40; WBCSD & WRI, n.d., 25–29).

The emissions discussed in this study are Scope 1 emissions from restaurants owned by Kampusravintolat Oy. Since LUT university owns and runs the campus restaurant together with LAB University of Applied Sciences (Kampusravintolat Oy, n.d.), emissions from things like ovens, stoves, fridges, and other on-site restaurant and kitchen equipment count as Scope 1. Electricity used in the restaurant is Scope 2, because it's bought from an energy provider. Emissions from transporting food and waste treatment off-site are Scope 3, since they happen outside the Kampusravintolat Oy's direct control. Packaging and kitchenware procurement are Scope 3 as they are purchased goods not made by the Kampusravintolat Oy.

Kampusravintolat Oy uses Poweresta for menu management (Mattila, S., 2024c). PoweResta CO₂ calculator for food services is developed in collaboration with Green Carbon Finland

Oy (PoweResta Oy, n.d.b). The CO₂ calculator is fully integrated into PoweResta kitchen management system allowing accurate, real-time tracking of the carbon footprint of each meal and communicating the climate impact of meals to customers via automated integration with the Mealdoo menu service (PoweResta Oy, n.d.a).

While this study uses carbon footprint calculations to estimate environmental impacts, it is important to acknowledge their limitations. As Finkbeiner (2009) mentions, carbon footprinting has a risk of oversimplifying complex environmental trade-offs. For instance, relying solely on CO₂ metrics leaves out broader sustainability issues such as water use, land degradation, or biodiversity loss. That is why it is important to interpret carbon data within a broader sustainability context and consider trade-offs that may occur when optimizing only for greenhouse gas emissions. According to Jolliet et al. (2015), carbon footprint is just one part of a life cycle assessment (LCA), and carbon footprint looks only at how much a product, activity, or company contributes to climate change through greenhouse gases. Additionally, when estimating the benefits of food waste prevention, the embodied and disposal costs of food waste could be considered with techniques such as Life Cycle Costing (LCC) and Environmental Impact Assessment (EIA-LCC) which can help to evaluate savings and trade-offs related to waste prevention (Chatterjee 2022, 126).

As a practical example, sustainability verification across supply chains was mentioned as a concern in the Food Waste Ecosystem Seminar. During the seminar presentation, A. Immonen from YLVA raised questions about the practical challenges of verifying sustainability claims made by suppliers, particularly in relation to traceability and the risk of greenwashing. These concerns were further reflected in YLVA's presentation, which emphasized the complexity of tracing the origins and assessing the environmental impacts of imported raw materials, especially for high-risk ingredients such as nuts, cocoa, and rice (Immonen, 2024).

The following chapter presents the results and findings, including calculated CO₂ emissions reduction estimates based on the demand forecasting solution piloted during this project. The calculations apply the carbon accounting principles discussed above to real-world food waste data collected at Kampusravintolat Oy.

5 Results and findings

This chapter presents the results of the Waste Watch project with an analysis of its environmental and operational impact. It includes quantitative findings on waste reduction and estimates associated CO₂ emissions avoided. In addition, the chapter discusses the broader practical relevance of the solution by connecting the results to global sustainability targets.

The chapter addresses Research Question 1 by evaluating how data-driven forecasting influenced food waste reduction and emission outcomes. It also responds to Research Question 2 through a detailed calculation of carbon footprint reductions achieved during the pilot period. Finally, the analysis of how the demand forecasting tool worked in practice, how staff used it, and how ready the institution was to adopt it helps to answer Research Question 3, also showing what conditions are needed to scale the solution for wider impact.

5.1 Quantitative reduction in food waste

To assess the climate impact of the Waste Watch, the project used a practical estimation approach based on real-world operational data and institutional emissions baselines. The goal was to quantify the preliminary CO₂-equivalent (CO₂e) reduction achieved by lowering service waste through improved demand forecasting.

The starting point for the calculation was the 2023 carbon footprint for LUT Buffet, which was reported to be 406 tons of CO₂-equivalents (CO₂e) for the year (LUT University 2024.)

Figure 3 shows the composition of total food waste generated at LUT Buffet between August 2023 and July 2024, totaling 9887 kilograms per year. The three largest categories are:

- Plate waste ('Lautasjäte') 44%: Edible food left uneaten by customers on their plates
- Service waste ('Linjastohävikki') 41%: Prepared food placed on the buffet line but not served
- Donated food ('Lahjoitus') 6%: Surplus food donated rather than discarded

The focus of this study is on service waste, as it directly results from overproduction and is therefore preventable through improved demand forecasting. Plate waste and donated food were excluded from the emissions analysis, as their reduction falls outside the scope of forecasting intervention.

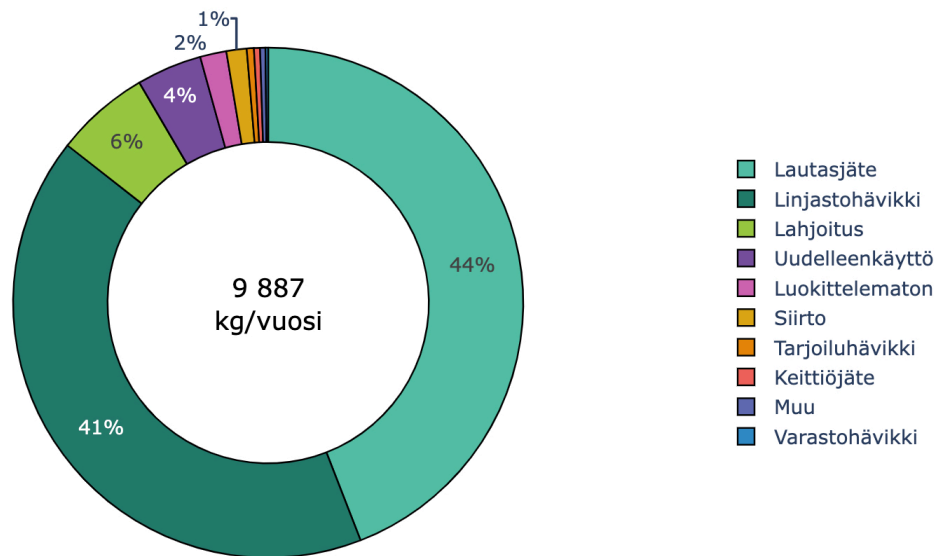


Figure 3. LUT Buffet's waste composition during Aug 2023 – July 2024 (Source: Waste Watch application)

Figure 4 is an example of how one week's diner amount prediction looks like in the software, showing daily diner amount predictions and actual diner numbers at LUT Buffet during weeks 19–21 of 2025, as generated by the Waste Watch application. The table includes the forecasted number of diners ('Ennuste'), actual diner counts ('Tänä vuonna'), the accuracy of the forecast ('Tarkkuus'), and the corresponding diner counts from the previous year ('Viime vuonna').

The forecast model performed with varying accuracy across the weeks shown on Figure 4. For example, on 6th of May 2025 (Tuesday), the prediction was 884 diners compared to the actual 800, resulting in a 9.5% overestimation. On 12th of May (Monday), the model overestimated traffic, predicting 827 diners while only 708 attended, resulting in a 14.4% error. However, in some cases, the model closely matched reality: on 8th May, it predicted 814 diners, and the actual count was 773, resulting in only a 5% deviation.

LUT Buffet ▾
vko 20 ▾
2025 ▾

Viikko	Päivä	Pvm	Ennuste	Tänä vuonna	Tarkkuus	Viime vuonna
19	Ma	2025-05-05	896	960	-7.2	845
	Ti	2025-05-06	884	800	9.5	914
	Ke	2025-05-07	894	936	-4.7	815
	To	2025-05-08	814	773	5	0
	Pe	2025-05-09	568	575	-1.3	0
20	Ma	2025-05-12	827	708	14.4	842
	Ti	2025-05-13	835	768	8	782
	Ke	2025-05-14	726	775	-6.8	766
	To	2025-05-15	653	626	4.2	615
	Pe	2025-05-16	468			581
21	Ma	2025-05-19	795			656
	Ti	2025-05-20	823			663
	Ke	2025-05-21	669			753
	To	2025-05-22	683			590
	Pe	2025-05-23	492			494

Figure 4. LUT Buffet's diner amount prediction example (Source: Waste Watch application)

These figures show that even though the forecasting tool is generally accurate, fluctuations in diner attendance can still result in forecast error, particularly due to external factors such as exams, events, or weather. The inclusion of historical data from 2024 gives additional context, showing that predicted values often align more closely with prior year trends. This shows the need for continuous forecasting model refinement and adding of other contextual variables to improve accuracy.

Ma (2024) writes about importance of interpreting relative error figures critically and preserving historical forecasts for long-term performance evaluation. Many restaurant forecasting models suffer from structural bias particularly when demand exceeds production capacity in central kitchens or buffet restaurants. In such cases excess demand often goes unrecorded and makes forecast accuracy appear better than it truly is. This can lead to a false sense of precision and overconfidence in model performance.

To address this issue, Ma (2024) proposes estimating unconstrained demand by modelling unmet demand and adjusting error metrics accordingly. While this study does not implement unconstrained demand estimation, it evaluates forecast accuracy based on observed diner counts, assuming all demand was met. This may understate forecast error in cases when production limits restricted meal availability.

A practical example of constrained demand concerns occurred in the “To the last crumb” (‘Viimeistä murua myöten’) model used by Sodexo at Stadin Ammattiopisto’s Vilppulantie unit in Helsinki, where production of certain meals is stopped during the last 30 minutes of lunch to minimize waste, and it has reduced service waste by 10 grams per portion (Mattila, 2024b). However, the staff reported concerns that families might provide negative feedback if their child did not receive a specific meal, and while such a model may be suitable in vocational settings where students are older, it may have greater challenges in contexts involving younger children, where expectations around meal availability can be more sensitive (Ibid.) However, in the context of university campus restaurant, unconstrained error is not an issue because as campus restaurants offer multiple dishes and the campus area has several restaurants.

Figure 5 shows the item-level menu forecasting view from the Waste Watch application for LUT Buffet on week number 22, starting on 26th of May 2025. The table displays individual menu items, the number of predicted diners per item (‘Ruokailija’), the corresponding weight in kilograms (‘kg’), the number of GastroNorm (GN) containers needed (‘Pakki’), actual containers prepared (‘Tehty’), portion unit size (‘Yksikkö’), leftovers (‘Jäi’), and any staff notes (‘Muistiinpanot’).

vko 22
2025
LUT Buffet
Print to PDF
Save

Pvm.		Ruokalaji	Ruokailija	kg	Pakki	Tehty	Yksikkö	Jäi	Muistiinpanot
26-05	KK	Kurpitsa-kookoskeittoa	47	18	2	4	1/2-150 ros. keitto, kastike	2	
	KL	Härkis-kaalilaatikkoa	77	31	4	7	1/1-100 ros. lasagne	2	
Rkm. Tot.	L1	Kalkkuna-kasvisnuudelit	83	46	5	7	1/1-100 ros. lasagne	2	14 kg liha
618	L2	Paistettu Kananmuna	465	28	6				
	L2	Pyttipannua	277	111	13	19	1/1-100 ros. lasagne	2	

Figure 5. LUT Buffet’s menu item prediction example (Source: Waste Watch application)

For example, the tool predicted 47 diners for the pumpkin-coconut soup ('Kurpitsa-kookoskeittoa'), translating to 18 kg of food and 2 GastroNorm containers. Staff prepared 4 containers, with 2 left over. Similarly, for fried egg ('Paistettu kananmuna'), 465 diners were predicted, requiring 28 kg and 6 containers, while 6 were made and 2 were left over. This detailed level of forecasting helps kitchen staff to prepare meals in appropriate quantities and make data-informed decisions about batch sizes and production adjustments.

The system also includes contextual notes, such as "14 kg meat" ('14 kg liha') under turkey-vegetable noodles ('Kalkkuna-kasvisnuudelit'), which may refer to staff member wanting to track how much meat was used for different reasons, such as deviations from the standard recipe due to ingredient availability. This figure illustrates how the forecasting tool integrates diner predictions, weight calculations, and production quantities into a practical interface that directly supports operational planning and food waste prevention in campus kitchens.

Figure 6 viewable on the next page shows the monthly service waste (in kilograms) recorded at LUT Buffet from August 2022 to May 2025, based on data from the Waste Watch application. The results show a downward trend in total annual service waste over the three-year period, decreasing from 5449 kg in 2022–2023 to 4135 kg in 2023–2024, and further to 4041 kg in 2024–2025. The most significant reductions occurred during the spring months, particularly in April and May. For example, service waste in April 2022–23 was 320 kg, compared to just 217 kg in April 2024–25.

Additionally in figure 6, seasonal variation is visible across all years, with peaks typically observed in September and January, coinciding with the time when new students arrive on campus, and some are introduced for the first time to the Finnish cuisine. The peak periods in September and January could be tackled by integrating food waste awareness as a part of the university-level sustainability training during the orientation week program. On the other hand, lower values occur during December and spring. The data suggest that the introduction and iterative development of the forecasting system contributed to a reduction in overproduction during the later months of the 2024–25 period.

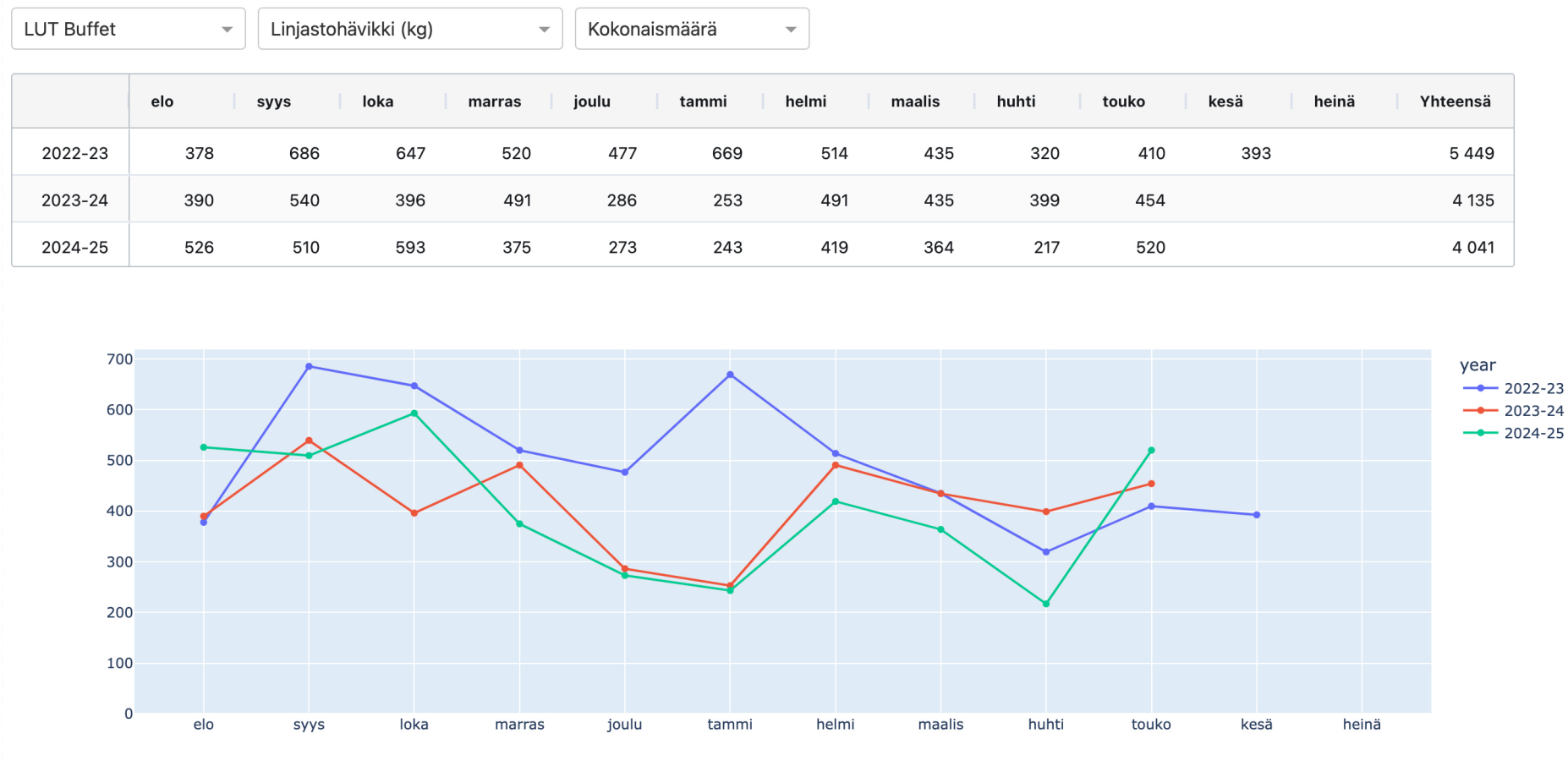


Figure 6. LUT Buffet's service waste during Aug 2022 – May 2025 (Source: Waste Watch application)

The Waste Watch forecasting system was deployed in March 2025. Waste measurement data from the restaurant's smart scale system BioVaaka and kitchen logs allowed comparison of service waste volumes for March, April, and May between 2024 and 2025. These three months were selected because they are from the period when the Waste Watch software was actively used in the kitchen, and also seasonal meal volumes remained relatively consistent according to the historical data. The resulting service waste reduction in kilograms was as documented in Table 5.

Table 5. Service waste reduction in kilograms

Month	2024 Waste (kg)	2025 Waste (kg)	Reduction (%)
March	435	364	-16%
April	399	217	-46%
May	454	520	+15% (increase)

The results in Table 5 show a notable decrease in service waste during March and April 2025 compared to the same months in 2024. The largest reduction occurred in April, with a 46% decrease in waste. In March, the waste decreased by 16%, also indicating improved production planning. However, in May 2025, service waste increased by 15% compared to the previous year. This spike may be explained by inaccuracies in the forecast due to end-of-semester fluctuations. Overall, the data indicates that the tool was particularly effective in reducing overproduction during the early spring months.

5.2 CO₂ emissions reduction

Due to variation in May, an average reduction rate of 16% across the three months was used as a conservative and preliminary estimate for potential long-term impact of the forecasting tool. The estimated emissions savings were calculated using formula 1 and values, summarized in table 6 below.

Table 6. Emission calculation datapoints

Variable	Value	Description
Total annual emissions	406 t CO ₂ e	Reported emissions from LUT Buffet (2023)
Share of service waste	47%	Portion of total emissions attributed to overproduction
Observed waste reduction	16%	Reduction in line waste during pilot period
Estimated CO ₂ e reduction	30.5 t CO ₂ e	Annual emissions avoided due to forecasting intervention

The calculation used the following formula to estimate annual CO₂e savings:

Formula 1: Annual emissions × Share of service waste × Waste reduction rate = Estimated CO₂e savings

Substituting the known values:

$$406 \text{ t CO}_2\text{e} \times 0.47 \times 0.16 = 30.5 \text{ t CO}_2\text{e}$$

This means that if the current reduction trend continues, the LUT Buffet restaurant alone could achieve an annual emissions reduction of approximately 30.5 metric tons of CO₂e by minimizing overproduction.

The project also developed a CO₂ emissions dashboard integrated into the Waste Watch application. This dashboard links large trays used for cooking and food display, production volumes, dish types, and average food category emission factors. While the calculation is not as detailed as item-specific analyses, this method enables scalable emissions accounting suitable for high-volume food service environments such as campus restaurants.

In summary, the emissions reduction estimate of 30.5 tons CO₂e per year is a step towards more sustainable institutional food service practices. Although these figures are preliminary and accuracy is expected to increase over time, the findings support the conclusion that predictive forecasting can reduce climate impact when applied to large-scale buffet production.

5.3 Operational impacts

The implementation of the Waste Watch software introduced several operational changes in the campus food services affecting the long-term sustainability impacts. These impacts, while not directly measured in quantitative terms, show important changes in kitchen practices, user perception, and waste handling routines.

Changes in kitchen workflow

One of the most immediate operational impacts was the change in routine kitchen practices. According to feedback from head chef R. Hynninen (Mattila 2025h), chefs have started checking the daily demand forecasts as part of their routines. Additionally, cooking and display tray measurements conducted during the project helped align the software's forecasting metrics with units used in actual kitchen operations. This is making it easier for the kitchen staff to estimate appropriate cooking volumes.

Staff and student perceptions

In addition to quantitative changes, several spontaneous comments were received from students noting the improved quality of meals during the spring period. While the feedback was not collected systematically as a part of this project and hereby cannot be used to draw scientific conclusions, this feedback may indicate positive side effects because of kitchen staff paying attention which types of foods result in least waste, and how recipe improvements can diminish food waste.

Improved food surplus use

During the project period, Kampusravintolat Oy increased two practices aligned with waste reduction. Leftover food sales were increased as an initiative to sell surplus food that was still safe for consumption, and increased food donations were organized in cooperation with local food aid groups. These activities help to find new ways to use edible food that might have otherwise become food waste.

Additionally, the Waste Watch software includes features to track monetary loss due to food waste and make cost-saving estimates through avoided overproduction. While the full financial impact analysis requires longer-term data, early figures suggest the tool can contribute to reducing procurement costs and improving inventory management.

5.4 Forecast accuracy, results verification and limitations

The results presented in this study are based on a combination of empirical data, forecast performance monitoring, and emissions estimation methods. While every effort was made to ensure reliability and transparency, several limitations and verification considerations must be acknowledged.

Forecast accuracy is crucial for the effectiveness of demand prediction systems. Throughout the project, forecast error was continuously monitored. At Restaurant Skinnarila, the system maintained a 20-day rolling median relative error below 10%, indicating high short-term reliability as shown in Figure 7.

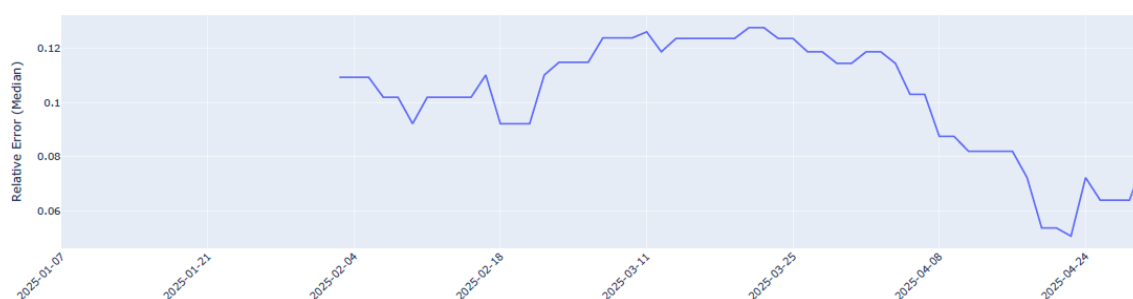


Figure 7. 20-day rolling median of relative forecasting error for Restaurant Skinnarila

Similarly, at LUT Buffet, forecast performance remained stable throughout the pilot period in spring 2025, with a mean relative error around 10%, as visualized in Figure 8. These metrics provide confidence that the software's forecasts were sufficiently accurate to guide kitchen decision-making.

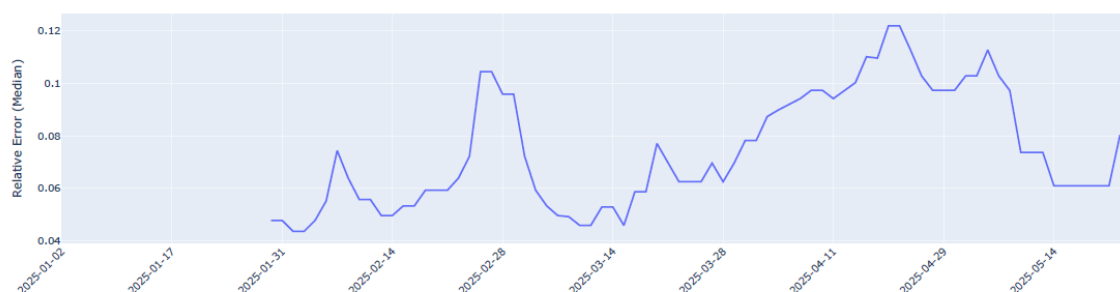


Figure 8. 20-day rolling median of relative forecasting error for LUT Buffet

Emission savings were estimated using a CO₂ calculator integrated to track the carbon footprint of each meal in the menu management system PoweResta, developed for food services in collaboration with Green Carbon Finland Oy (PoweResta Oy, n.d.a; PoweResta Oy, n.d.b). As a critical reflection, the emission factors applied by the carbon footprint calculator show average climate impacts per kilogram of food for different food categories, but these are approximate averages and do not capture the full diversity of ingredient combinations or regional sourcing. The carbon footprinting data obtained via PoweResta were applied to aggregate production volume data linked to kitchen trays. The project team had no access to detailed recipes, which the restaurant considered proprietary business information which could not be shared. While lack of detailed recipes limits detail, the overall method of using representative food categories to estimate average carbon footprints of meal portions is in accordance with practices described in institutional-level LCA studies such as Virtanen et al. (2011).

At the time of writing, third-party audit of the emission estimates has not been conducted. Verification was limited to internal checks on the consistency of calculator inputs and alignment with documented cooking volumes. This shows a broader challenge in food service decarbonisation where transparent, verifiable CO₂ metrics are still emerging, and many practitioners rely on modelled averages or best-available proxies.

When connecting food waste reduction to measurable environmental benefits, this study relies on assumptions based on average standard emission factors used by PoweResta's carbon footprint calculator, rather than detailed, site-specific life cycle analyses. This introduces a degree of uncertainty because as shown in the literature, emission factors can vary substantially depending on the type of food, production method, and region. For example, Virtanen et al. (2011) found that emissions from red meat can be more than ten times higher per kilogram than those from plant-based foods, making visible the impact of food type on carbon footprint. Additionally, many studies do not clearly document their calculation methods, making it difficult to compare results across different cases (Virtanen et al., 2011).

In this project, the emissions reduction estimates should be viewed as indicative rather than precise measurements. The objective was not to produce a definitive carbon audit but rather to assess the size and direction of impact, using a method backed up by recognized scientific sources.

Finally, the role of human behaviour and institutional culture is both an opportunity and a limitation. The forecasting model does not directly account for behavioural factors such as portion sizes, student preferences, or kitchen staff improvisation, even though these factors influence actual food waste. While stakeholder engagement was a part of the intervention process, its role in influencing outcomes was not quantitatively measured. Future research could study the integration of behavioural insights into forecasting tools, as well as the long-term effects of such interventions on institutional practices.

6 Discussion

This research project examined whether data-driven demand forecasting can effectively reduce food waste and associated CO₂ emissions in university campus restaurants. The results indicate that integrating predictive tools into kitchen workflows can lead to reductions in food waste and help institutions align daily operations with climate goals. These findings are relevant for institutional kitchens seeking scalable strategies for waste prevention.

The pilot implementation of the Waste Watch tool at LUT University's Lappeenranta campus demonstrated an average 16% reduction in service waste over a three-month period, corresponding to an estimated 30.5 metric tons of avoided CO₂-equivalent emissions annually. While results varied from one month to another, they consistently showed that demand forecasting can reduce overproduction.

George et al. (2021) define digital sustainability as the use of digital innovations to tackle climate and sustainability challenges in a way that enhances resource efficiency and avoids unintended harm. The forecasting tool developed in this project uses existing data streams and lightweight software infrastructure, avoiding energy-intensive technologies while delivering outcomes that improve sustainability.

As George et al. (2021) state, digital sustainability innovations operate at the intersection of technology, entrepreneurship, and systems transformation, and are most impactful when they address complex environmental challenges, such as climate change or resource inefficiency while being embedded in context-specific institutional processes. This project has such an approach by enabling real-time adjustments in food production practices through accessible digital means within an institutional kitchen context. In summary, this project shows how a modest but well-integrated digital intervention can support sustainability goals in the food services.

Operational feedback was that kitchen staff incorporated the daily forecasts into their production planning, with forecasted tray quantities increasingly complementing habit-based estimations. The tray-based design and real-time integration of menu and point-of-sale (POS) data helped to assure practicality, relevance to daily operations, and ease of adoption. These features differentiate Waste Watch from traditional post-consumption tracking tools and offer a proactive approach to food waste prevention.

When compared to previous initiatives such as the University of Vaasa's NextGen project, Waste Watch emphasizes the importance of preventing overproduction before meals are cooked, rather than tracking waste after the production. Waste Watch system has a lean software architecture which allows fast and low-cost deployment while NextGen relied on full-stack Azure development with higher software development overhead. Despite initial concerns about vendor costs for data integration that occurred in earlier food waste reduction projects, the Waste Watch team secured free or low-cost API access in most cases. This suggests that negotiation with vendors in the context of institutional climate action can make a difference in cost-effective implementation.

However, API implementation was not without challenges as a critical delay was caused by one vendor's slow API delivery, which postponed automated data flows. This experience brought up recurring issues in public sector software procurement, including fragmented infrastructure and slow system integration. To address this, the project launched a voluntary initiative advocating for open or low-cost API access from vendors to support sustainability innovation.

The implementation of a data-driven forecasting tool in a real-life food service environment raised some ethical considerations. While the goal of the project is to reduce food waste and associated emissions, it is important to also critically reflect on the broader effects of automation, data collection, and ethics.

One ethical concern is the potential deskilling or disempowerment of staff in institutional kitchens. Even though predictive systems can increasingly inform production volumes and menu planning, rather than replacing human judgement, forecasting tools should be positioned as decision-support systems. Ensuring staff are involved in interpreting and acting upon forecasts is important for ownership and agency in the workplace.

Another concern involves data governance. Although the data used in this study were collected from operational systems and waste monitoring devices were already in use, the ethical use of such data requires attention to transparency, consent, and privacy. Even in organizational settings, questions arise about who controls the data, how it is used, and whether it might be repurposed beyond its original purpose. In this project, these concerns were considered when making decisions about the software infrastructure and data storage.

This project supported progress toward several UN Sustainable Development Goals (SDGs) related to sustainable urban systems, responsible resource use, and climate action.

Goal 11: Sustainable Cities and Communities

(Target 11.6: Reduce the adverse environmental impact of cities, with particular attention to municipal and other waste management)

EU waste hierarchy prioritizes prevention as the most desirable action in waste reduction. In alignment with this, the Waste Watch project contributed to upstream food waste reduction in an urban university context. By optimizing food preparation using forecasting tools, the initiative reduced the volume of organic waste generated. Consequently, this eases the pressure on waste treatment infrastructure and supports more sustainable urban waste management.

Goal 12: Responsible Consumption and Production

(Target 12.3: Halve per capita global food waste at the retail and consumer levels, and reduce food losses along production and supply chains)

The forecasting model helps the restaurant to more accurately estimate the number of expected diners. Adjusting meal production accordingly is reducing the likelihood of surplus. At the same time with this study, a parallel thesis focused on plate waste and identified behavioural and operational factors contributing to it. Together, these efforts improve food use efficiency across both production and consumption stages.

Goal 13: Climate Action

(Target 13.3: Improve education, awareness, and institutional capacity on climate change mitigation and adaptation)

With the development and deployment of a demand forecasting solution, the project enhanced institutional capacity to monitor and reduce food-related CO₂ emissions. It also contributed to staff competence and awareness, with concrete changes in kitchen practices such as recipe optimization, portion adjustments, and the increase of surplus food sales and food donations, reflecting an increased commitment to climate-conscious operations.

The project's broader applicability is promising, especially within networks of institutional kitchens that already utilize digital waste tracking and smart scales. In terms of policy and education, the findings support the inclusion of forecasting systems in institutional sustainability strategies. At LUT University, results may guide future emissions reduction measures, such as incorporating food waste literacy into student orientation or expanding forecasting software use across other campuses.

Several limitations should be acknowledged. The forecasting tool was tested only at two restaurants within a single campus, and the CO₂ calculations relied on average emission factors used by menu management system's carbon footprint calculator, rather than detailed life-cycle assessments. Forecast error was not corrected for unmet demand, and external verification of results is still pending. These issues suggest the need for caution in generalizing findings across different institutional contexts.

Nonetheless, the pilot revealed multiple avenues for future research and development such as piloting the solution in new environments (e.g., staff restaurants or other university campuses), integration of plate waste monitoring and behavioural nudges, commercialization as Software-as-a-Service (SaaS) or as kitchen system integration, and creation of open-access datasets for replication and benchmarking.

7 Conclusions

This study researched the role of data-driven demand forecasting in reducing food waste and CO₂ emissions in university campus restaurants. Using an action research methodology, the study piloted and evaluated the Waste Watch forecasting tool in cooperation with Kampusravintolat Oy at LUT University's Lappeenranta campus.

The research addressed three main questions concerning the potential of demand forecasting in food services: reduction in food waste and CO₂ emissions, types of reductions and environmental impact, and implementation enablers and barriers.

The findings show that predictive forecasting tools, when effectively integrated, can reduce service waste by an average of 16%, equating to an estimated 30.5 tons of avoided CO₂ emissions annually. These results are consistent with the goals of national and international sustainability initiatives and show the potential of machine learning based tools to support institutional emissions reduction practices, particularly by reducing overproduction in high-emission meal categories.

The Waste Watch system also improved operational planning, supported staff decision-making, and sparked new sustainability actions such as surplus food sales and increased food aid donations. Its development process revealed critical success factors such as early stakeholder involvement, user-friendly design, and ongoing iteration. The main challenges included delays in vendor data access, adapting the solution for the pilot restaurants workflows, and the initial costs of system development and integration.

This study contributes to the emerging field of machine learning-based forecasting as a practical climate intervention in institutional food services. It has both technical and organizational insights that could be useful for future deployments in similar environments. Additionally, the project supports institutional progress toward the UN Sustainable Development Goals 11 (sustainable cities), 12 (responsible consumption), and 13 (climate action).

While promising, the findings are specific to the local context and should not be generalized without further validation. The findings cannot be generalized without caution because food waste generation patterns, staff readiness for implementing new practices, and digital

readiness vary across institutions. Also, emissions estimates were based on assumptions and could be further refined with more detailed data.

This research contributes to the academic and practical understanding of digital sustainability interventions in food services. It is among the first documented use cases of machine learning-based demand forecasting in a Finnish university restaurant, offering insights into emissions tracking, operational behaviour, and software development. The system's real-time API integration and tray-based estimations represent methodological advancements over more reactive Excel-based waste tracking approaches.

In conclusion, the research demonstrates that proactive, data-informed decision-making offers measurable environmental and operational benefits in campus restaurants. Demand forecasting tools can act as enablers of institutional climate action and behaviour change. With continued use, refinement, and wider deployment, such systems could become a cost-effective, scalable component of sustainable food service models in higher education and beyond.

References

- Alhonnoro, L., 2023a. *Buffetruokailun hävikki haltuun*. Recording of presentation at Tiedekarnevaalit. Available at: <https://www.youtube.com/watch?v=bWythvH9Vbo>
- Alhonnoro, L., 2023b. *Innovative solutions to reduce food waste in restaurants*. Waste Insight, SC-Research / University of Vaasa. Presented at Innovative Food Business, Seinäjoki, 6 September 2023. Available at: <https://sites.uwasa.fi/wasteinsight/wp-content/blogs.dir/4/files/sites/151/2023/11/Innovative-Solutions-to-Reduce-Food-Waste-in-Restaurants.pdf>
- Chatterjee, P., (ed.), 2022. *Machine learning algorithms for engineering applications: future trends and research directions*. New York: Nova Science Publishers.
- Clark, Q.M., Kanavikar, D.B., Clark, J. and Donnelly, P.J., 2024. *Exploring the potential of AI-driven food waste management strategies used in the hospitality industry for application in household settings*. *Frontiers in Artificial Intelligence*, 7, Article 1429477.
- EUR-Lex., 2024. *Waste hierarchy*. Available at: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=LEGISSUM:waste_hierarchy
- FAO, 2025a. *Food waste*. Food and Agriculture Organization of the United Nations. Available at: <https://www.fao.org/platform-food-loss-waste/food-waste/introduction/en>
- FAO, 2025b. *Food loss*. Food and Agriculture Organization of the United Nations. Available at: <https://www.fao.org/platform-food-loss-waste/food-loss/introduction/en>
- Finkbeiner, M., 2009. *Carbon footprinting—opportunities and threats*. *The International Journal of Life Cycle Assessment*, 14(2), pp. 91–94. Available at: <https://doi.org/10.1007/s11367-009-0064-x>.
- Finnish Hospitality Association MaRa, 2021. *Reducing food waste in restaurants*. Available at: <https://www.mara.fi/media/vastuullisuus/reducing-food-waste-mara.pdf>
- George, G., Merrill, R.K. and Schillebeeckx, S.J.D., 2021. *Digital sustainability and entrepreneurship: How digital innovations are helping tackle climate change and sustainable development*. *Entrepreneurship Theory and Practice*, 45(5), pp. 999–1027.
- Hukka AI Oy, 2025. *Tuote – Hukka AI*. Available at: <https://hukka.ai/tuote/>
- Hyvärinen, H., 2024. *Selvitys ravitsemispalveluiden ruokahävikin vähentämiseen liittyvistä haasteista ja mahdollisuuksista*. Kaupunkiympäristön aineistoja 2024:10. Helsingin kaupunki / Kaupunkiympäristön toimiala. Available at:

<https://www.hel.fi/static/liitteet/kaupunkiymparisto/julkaisut/aineistot/aineistoja-10-24.pdf>

Hübner, N., Caspers, J., Coroamă, V.C. and Finkbeiner, M., 2024. *Machine-learning-based demand forecasting against food waste: Life cycle environmental impacts and benefits of a bakery case study*. Journal of Industrial Ecology, 28(5), pp. 1117–1131.

IEA, 2025. *Energy and AI*. Paris: International Energy Agency. Available at: <https://www.iea.org/reports/energy-and-ai>

Immonen, A., 2024. *Vastuullisuus ja hävikin vähentäminen UniCafe-ravintoloissa*. Presentation at the Food Waste Ecosystem Hävikkiviikko seminar, 27 September 2024. Unpublished presentation slides received via seminar communication.

Jolliet, O., Saade-Sbeih, M., Shaked, S., Jolliet, A. and Crettaz, P., 2015. *Environmental life cycle assessment*. 1st ed. Boca Raton: CRC Press. Available at: <https://doi.org/10.1201/b19138>

Kampusravintolat Oy, 2025. *Yhteystiedot*. Available at: <https://www.kampusravintolat.fi/yhteystiedot/>

Kananen, H. and Puolitaival, H., 2019. *Tekoäly: bisneksen uudet työkalut*. Helsinki: Alma Talent Oy.

Kanwal, N., Zhang, M., Zeb, M., Batool, U., Khan, I., Rui, L. and others, 2024. *From plate to palate: Sustainable solutions for upcycling food waste in restaurants and catering*. Trends in Food Science & Technology, 152, p.104687.

Knorr, D. and Augustin, M.A., 2024. *From food to gods to food to waste*. Critical Reviews in Food Science and Nutrition, 64(16), pp. 5379–5397.

Kuusisto, A., 2025. *NextGen project presentation*. University of Vaasa. Presented at meeting of LUT University, University of Vaasa and Biovaaka Oy at Lappeenranta, 27 May 2025. Unpublished presentation slides.

Leanpath, Inc., 2025. *Food waste management solutions*. Available at: <https://www.leanpath.com/food-waste-management-solutions/>

Natural Resources Institute Finland (LUKE), n.d. *Volume of food loss and waste*. Available at: <https://www.luke.fi/en/statistics/indicators/finlands-cap-impact-indicators/volume-of-food-loss-and-waste>

New Gastroline GmbH, 2025. *GN Behälter*. Available at: <https://www.gastrodax.de/gastrobedarf/gn-behaelter>

Ma, B., Cui, H., Yan, L., Zhang, Y., Wang, J. and Zhang, Y., 2023. *Analyzing research trends of universities' carbon footprint: An integrated review*. Gondwana Research, 121, pp. 259–275.

- Ma, J., 2024. *Estimating restaurants' unconstrained demand: A systematic approach to reducing structural bias in forecast accuracy measures*. *Journal of Hospitality and Tourism Technology*, 15(3), pp. 363–378.
- Mattila, S., 2024a. *Food Waste Ecosystem Hävikkiviikon ruokahävikkiseminaari, various speakers*, 27 September 2024. Unpublished seminar notes.
- Mattila, S., 2024b. *Food Waste Ecosystem: Best practices in donating surplus food*. 18 April 2024. Unpublished seminar notes.
- Mattila, S., 2024c. *Meeting with L. Malinen, Kampusravintolat Oy*, 16 August 2024. Unpublished meeting notes.
- Mattila, S., 2024d. *Meeting with L. Malinen and J. Hänninen*. 12 June 2024. Unpublished meeting notes.
- Mattila, S., 2024e. *Notes from software benchmarking meeting with University of Vaasa's NextGen/Wasteless predictive modelling project team and Biovaaka software developer*. 27 August 2024. Unpublished meeting notes.
- Mattila, S., 2024f. *Notes from presentation by J. Laine on software architecture and procurement at Helsinki Incubators by University of Helsinki*. Unpublished learning notes.
- Mattila, S., 2024g. *Meeting with J. Vänskä at LUT Lahti campus*, 23 May 2024. Unpublished meeting notes.
- Mattila, S., 2024h. *Meeting with R. Hynninen at LUT Lappeenranta campus*, 27 May 2025. Unpublished meeting notes.
- Mattila, S. and Sharbaf, J., 2025. *Meeting with L. Malinen & R. Hynninen at Kampusravintolat Oy*, 6 February 2025. Lappeenranta. Unpublished meeting notes.
- McNiff, J., 2013. *Action Research: Principles and Practice*. 3rd ed. Milton Park, Abingdon, Oxon: Routledge.
- Menzies, T. et al. *Powering down: An interview with Federica Sarro on tackling energy consumption in AI-powered software systems*. *IEEE Software*, 41(5), pp. 89–92.
- Luonnonvarakeskus & OSAO. 13.3.2025. *Hävikkihankkeiden pyöreä pöytä*. Seminar. Helsinki.
- LUT University, 2024. *Sustainability Report 2023*. Lappeenranta: LUT University. Available at: <https://www.lut.fi/sites/default/files/media/documents/LUT-University-Sustainability-Report-2023.pdf>
- O'Brien, R. 2001. *An Overview of the Methodological Approach of Action Research*. In: R. Richardson, ed. *Theory and Practice of Action Research*. Available at: <http://www.web.ca/~robrien/papers/arfinal.html>

PoweResta Oy, n.d.a. *Hiilijalanjäljen laskenta ravintolassa*. Available at:

<https://poweresta.com/hiilijalanjalki-ravintola-alalla/>

PoweResta Oy, n.d.b. *PoweRestan uusi CO₂-laskuri muuttaa ravintoloiden ja ammattikeittiöiden hiilijalanjäljen hallinnan ainutlaatuisen datan avulla*. Available at:

<https://poweresta.com/liiketoiminta/powerestan-uusi-co2-laskuri-muuttaa-ravintoloiden-ja-ammattikeittioiden-hiilijalanjaljen-hallinnan-ainutlaatuisen-datan-avulla/>

Riipi, I., Hartikainen, H., Silvennoinen, K., Joensuu, K., Vahvaselkä, M., Kuisma, M. & Katajajuuri, J-M., 2021. *Elintarvikejätteen ja ruokahävikin seurantarjestelmän rakentaminen ja ruokahävikkitiekartta*. Luonnonvara- ja biotalouden tutkimus 49/2021. Luonnonvarakeskus. Helsinki. Available at:

<https://jukuri.luke.fi/handle/10024/547657>

Ritchie, H., 2020. *Food waste is responsible for 6% of global greenhouse gas emissions*.

Our World in Data. Available at: <https://ourworldindata.org/food-waste-emissions>

Ronen, K.I., 2019. *Action Research as a Methodology for Professional Development in Leading an Educational Process*. Studies in Educational Evaluation, 64, 100826.

Sahota, A. (ed.), 2014. *Sustainability: How the Cosmetics Industry is Greening Up*. 1st ed. Chichester: Wiley.

Sharbaf, J., 2025. *Waste Watch: Data-Driven Forecasting and Food Waste Reduction in Campus Restaurants*. Presentation at: Hävikkihankkeiden pyöreä pöytä, Luonnonvarakeskus & OSAO. 13 March 2025. Helsinki.

Silvennoinen, K. 10.6.2024. Senior Researcher, Luonnonvarakeskus. *Ravintoloissa syntyvä elintarvikejäte ja jätelain velvoitteet*. Vantaan kaupunki. Seminar. Vantaa.

Silvennoinen, K., 2021. *Työkalu ruokahävikin mittaamiseen ravitsemispalveluille: Lukeloki*. Natural Resources Institute Finland (Luke). Available at:

https://projects.luke.fi/ravintolafoorumi/wp-content/uploads/sites/4/2021/09/Lukeloki_esittely_foorumi_2021.pdf

Sodexo, 2024. *Pilottihanke vähentää ruokahävikkiä*. Presentation at the Food Waste Ecosystem Hävikkiviikko seminar, 27 September 2024. Unpublished presentation slides received via seminar communication.

Soupster, 2024. *Kokemuksia ruokahävikin mittaamisesta: Lukeloki*. Presentation at the Food Waste Ecosystem Hävikkiviikko seminar, 27 September 2024. Unpublished presentation slides received via seminar communication.

United Nations Environment Programme (2024) *Food Waste Index Report 2024. Think Eat Save: Tracking Progress to Halve Global Food Waste*. Available at:

<https://wedocs.unep.org/20.500.11822/45230>

Virtanen, Y., Kurppa, S., Saarinen, M., Katajajuuri, J.M., Usva, K., Mäenpää, I., Mäkelä, J., Grönroos, J. and Nissinen, A., 2011. *Carbon footprint of food – approaches from national input–output statistics and a LCA of a food portion*. Journal of Cleaner Production, 19(16), pp.1849–1856.

World Business Council for Sustainable Development WBCSD & World Resources Institute WRI, n.d. *The Greenhouse Gas Protocol: The GHG Protocol Corporate Accounting and Reporting Standard. Revised Edition*. Available at: <https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf>

Appendix 1. Benchmarking: Food waste technology solutions

Name	Country	Type of Technology	Status/Notes
Winnow	UK	Camera and scale system	Widely used globally
Leanpath	US	Camera and scale system	Integrated by Sodexo globally
Biovaaka	Finland	Smart scales, waste data collection	Focus on real-time monitoring and data reporting.
SmartKitchen	Finland	Smart scales, waste tracking, food safety software	Used in professional kitchens to optimize operations
Hukka AI	Finland	Reduce food waste via logging, measuring and analysing. Rentable smart scales.	For data-driven measuring, analysing and reducing food waste
NextGen Forecasting Model 3.0	Finland	Forecasting waste	Developed by University of Vaasa
Zipli	Finland	B2B food surplus distribution and sales	Early-stage startup
Tenzo	United Kingdom	Sales forecasting for restaurants	Uses data from POS systems, available for certain POS systems only
CGI Hävikkimestari	Finland	Integrated with CGI Aromi restaurant management system	Developed by Lassila & Tikanoja, acquired by CGI
Sensire	Finland	Food safety software	Quality and compliance management
Antell	Finland	Matches food quantities to demand and creates menus with AI	Proprietary software created in collaboration with software company Silo AI
Lukeloki	Finland	Waste tracking for restaurants	Created by Natural Resources Institute Finland (Luke)
Code Ananas	Finland	AI-powered mobile app that helps to reduce food waste based on photos of fridge contents and receipts	Early-stage startup

Appendix 2. Summary of stakeholder meetings conducted for the thesis

Interviewee(s)	Role/Organization	Date	Key Topics Discussed
J. Vänskä	Development manager, University of Vaasa	23.5.2024	Project advisory meeting, learning about University of Vaasa's former food waste forecasting project
P. Herrala	Restaurant Manager, Food&Co Isku Center, restaurant on LUT Lahti campus	27.5.2024	Kitchen visit. Kitchen and waste related operations, interest in becoming a pilot restaurant in the project
O. Luukkonen	Päijät-Hämeen Ateriapalvelut	24.5.2024	Päijät-Häme food services' experiences using Hukka AI for food waste tracking and prevention
L. Malinen & J. Hänninen	Restaurant Manager Kampusravintolat Oy & CEO at Biovaaka Oy	13.6.2024	Presenting Waste Watch project, interest in becoming a pilot restaurant in the project
J. Kesälahti	Hukka AI	26.6.2024	Hukka AI software for food waste reduction
M. Verkasalo	SmartKitchen	2.7.2024	SmartKitchen software and hardware for food waste reduction
J. Vänskä	Development manager, University of Vaasa	9.8.2024	Project advisory meeting
L. Malinen	Restaurant Manager Kampusravintolat Oy	16.8.2024	Waste Watch project collaboration
J. Vänskä, & J. Nieminen	University of Vaasa	27.8.2024	Project advisory meeting with University of Vaasa's forecasting project team
J. Hänninen	CEO, Biovaaka Oy	4.9.2024	Biovaaka software and API implementation
T. Laamanen	Poweresta	12.9.2024	Poweresta menu management system
T. Laamanen	Poweresta	18.9.2024	Poweresta API implementation
L. Malinen	Restaurant Manager Kampusravintolat Oy	31.10.2024	Kitchen visit at LUT Buffet restaurant
E. Karppinen	Restaurant industry professional	7.11.2024	Restaurant software insights from point of view of someone who has worked as a chef, restaurant manager and marketer
E. Ahola	Confirma Finland	19.11.2024	API implementation
I. Hämäläinen	Aalef Oy, restaurant on LUT Lappeenranta campus	28.11.2024	Food waste tracking and reduction practices in restaurant Aalef at LUT Lappeenranta campus
L. Malinen & R. Hynninen	Restaurant Manager & Head Chef, Restaurant LUT Buffet, Kampusravintolat Oy	11.12.2024	Discussing software development and kitchen practices

S. Liukkonen	Aalef Oy, Restaurant YOLO	13.1.2025	Interest in piloting WasteWatch software at their restaurant Aalef at LUT Lappeenranta campus
L. Malinen & R. Hynninen	Restaurant Manager & Head Chef, Restaurant LUT Buffet, Kampusravintolat Oy	6.2.2025	Production planning, forecasting use, partial preparation, food preferences, Biovaaka data use, tray-level decisions
E. Pusenius	Restaurant Manager, Restaurant Skinnarila by Kampusravintolat Oy	28.3.2025	Offered Waste Watch software to be used by Restaurant Skinnarila. Offer accepted.
L. Malinen & R. Hynninen	Restaurant Manager & Head Chef, Restaurant LUT Buffet, Kampusravintolat Oy	28.3.2025	Latest experiences using the software
L. Malinen & R. Hynninen	Restaurant Manager & Head Chef, Restaurant LUT Buffet, Kampusravintolat Oy	13.4.2025	Latest experiences using the software
R. Hynninen	Head Chef, Kampusravintolat Oy	27.5.2025	Usefulness of the forecast, latest experiences

Appendix 3. AI disclosure

OpenAI's ChatGPT was used during the thesis process to support ideation, outline structuring, and language refinement. The tool was also used to improve clarity, academic tone, and grammatical correctness in various parts of the text. The tool was used selectively and critically with all core analysis, argumentation, and conclusions produced by the author. The use of AI was limited to language and productivity support and does not replace original academic work or judgement.